

Pediatrics and child health OSCE file

Neonatology

Classification of Newborns

1. Gestational Age:

- Preterm (<37wks)
- Term (37-42wks)
- Postterm (>42wks)

2. Birth Weight:

- Extremely low birth weight : < 1000 grams
- Very low birth weight: 1000gm-1499gms
- Low Birth weight: 1500gm-2499gm
- Normal BW: 2500-3999gm
- Macrosomia: ≥ 4000 gm

3. BW/Length/HC VS GA:

- SGA: <10th centile of Lubchenco
- AGA: b/n 10th-90th centile
- LGA: above 90th centile



APGAR Score



The Apgar score rates:

Respiration, crying

Reflexes, irritability

Pulse, heart rate

Skin color of body
and extremities

Muscle tone

The APGAR Score

Sign	0	1	2
Heart rate	Absent	Slow (<100 bpm)	≥ 100 bpm
Respirations	Absent	Weak cry, Hypo-ventilation	Good, strong cry
Muscle Tone	Limp	Some flexion	Active motion
Reflex Irritability	No response	Grimace	Cough or sneeze
Color	Blue or pale	Body pink, Extremities blue	Completely pink

Essential new born care

Essential newborn Care

Essential newborn care is care given to all newborn infants at birth to optimize their chances of survival

- Step 1: Dry and Stimulate
- Step 2: Evaluate Breathing
- Step 3: Cord care
- Step 4: Keep the newborn warm (Prevent Hypothermia)
- Step 5: Initiate breastfeeding in the first one hour
- Step 6: Administer eye drops/eye ointment
- Step 7: Administer vitamin K Intramuscularly (IM)
- Step 8: Place the newborn's identification bands on the wrist and ankle
- Step 9: Weigh
- Step 10: Document

HEALTHY NEWBORN CARE

Care in delivery room

- *Resuscitate if required as per NRP guidelines with the ALS team*
- **Routine care:** if the baby has cried immediately after birth:
 - Place the baby on the mother's abdomen on pre-warmed dry and clean cloth. Dry the baby completely, changes the wet linen. Assess respiration and heart rate.
 - **Delayed cord clamping** at 1-3 minutes in both term and preterm neonates who do not require resuscitation. Cut the cord at 3-4 cm from skin, after applying plastic sterile clamp. Do not apply any medication.
 - **Skin to skin contact** with the mother. Place the baby prone on the mother's chest after covering the head and the body with a warm sterile sheet. Continue skin to skin contact for 1 hour. Even if the baby is separated for routines such as weighing place the baby in skin to skin contact. Ensure the safety of the baby continuously during the procedure.
 - **Early breastfeeding:** Help the mother for breastfeeding within 1/2 an hour of normal delivery in the labor room itself and in case of LSCS (low segment caesarian section) as soon as mother is out of anesthesia or within 4 hours of birth.
 - **Put identification tag**
 - Mother's name, sex, wt., delivery date and time should be included
 - For twins indicate which is first and second (follow the same for triplets and above)
 - **Examine the baby from head to toe:** Look for life threatening congenital anomalies like bilateral choanal atresia, esophageal atresia, tracheo-esophageal fistula, imperforate anus and diaphragmatic hernia. Do not forget to examine
 - a. Femoral pulsations.
 - b. Umbilical cord for number of arteries and veins.
 - c. Eyes for corneal size, corneal opacities and cataract.
 - d. Back of baby for tuft of hair, swelling and pilonidal sinus.
 - e. Placenta, specially in high risk mothers and neonates. Send for HPE(histopathology examination)in high risk cases.
- Assign sex to the baby and show baby to parents/ relatives.
- Weigh the baby, measure length & head circumference (after 24 hours) of the baby
- Give Inj. Vit.K 1mg IM to babies > 1500 g and 0.5 mg IM to babies < 1500 g.
- Apply TTC Eye ointment to both eyes to all babies.

- Collect cord blood for blood grouping in all cases.
- If the mother is O or Rh negative group, send for direct Coomb's test. Follow up DCT within 2 hours. Also do cord Hb and bilirubin if Rh negative mother.
- Cord ABG in babies requiring positive pressure ventilation, and high risk deliveries.

Care in PNC ward

- Keep mother & baby together "rooming in", "Bedding in" is preferred if the mother is conscious and able to care for the baby.
- Keep the baby warm by covering head, body and feet with cotton clothes.
- Monitor the temperature and feeding 6 hourly
- Feed the baby on demand; ensure proper position for breastfeeding & advice regarding breastfeeding. All mothers require lactation support. Assess for risk for lactation problems. At least one feed should be observed by the nurses and pediatrics resident doctor for proper position and attachment.
- Confirm whether the baby has passed meconium & urine within 1st 24 hrs and 48 hrs respectively.
- Check the weight daily. Watch for excessive (>10%) weight loss
- Assess risk factors for severe hyperbilirubemia. Examine the baby twice a day for jaundice. If the baby is icteric on day 1 or below abdomen at any time send for a TB/CB (total bilirubin/conjugated bilirubin) urgently.
- Supplement preterm newborns(less than 37 weeks) with Vitamin D3 400 IU.

If baby has not passed meconium in 24 hours

- Check for vomiting, abdominal distension to rule out surgical etiology.
- If absent - ensure proper supervised breastfeeding and re-examine after 2-4 hrs and reassure the mother.

If baby has not passed urine between first 12 to 24 hours

- Check the antenatal history for oligohydramnios and any renal malformations.
- Examine for signs of dehydration and a palpable bladder.
- Supervise feeds and re-examine after 4-6 hrs.

Skin care

- Don't remove vernix caseosa.
- **No baby bath immediately after birth.**
- No baby bath in hospital
- Examine daily for superficial skin infection.

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- Clean cord care – Do not apply anything over the cord and let the cord remain open to dry.
- Ensure cord is out of the diaper
- Watch for foul smell, periumbilical erythema or induration.

Eye care

- No special care required.
- Use Azithromycin syrup 10mg /kg for 5 days or ceftriaxone 125mg/kg STAS dose only if discharge is present, after sending discharge swab for gram stain and culture & sensitivity.
- Clean the eye with saline swabs from medial to lateral canthus before instilling eye drops after proper hand-wash.
- Apply TTC Eye ointment to both eyes to all babies.

Newborn Screening

- Collect cord blood for TSH-when available
- Hearing screening by behavioral observation audiometry
- OAE for high risk infants.
- Eye screening: by red reflex assessment
- Critical CHD screening

Management of LBW baby in the OB ward

- Manage LBW babies > 2000 g and > 34 weeks in the OB ward.(consult the ALS team and the pediatrics resident)
- Educate the family regarding keeping the baby warm and frequent feeding. Provide a warm cradle to the baby.
- Kangaroo mother care should be started immediately after delivery. If the mother is unable to do so, anyone from the family can provide KMC.
- Monitor temperature and feeding at least 4 hourly. The mother-baby dyad requires more intensive lactation support.
- Do RBS at 2 hours of birth and subsequently 6-8 hourly for the first 48-72 hours.
- Monitor for jaundice. Routinely test for TB /CB at 48 hours.
- Supplement with calcium, phosphorus and multivitamins.

Discharge policy

- Discharge full-term normal babies after 24 hrs & low birth weight babies after 3 days of postnatal life.

Examination on discharge

- Weight
- Presence of cardiac murmur.
- Umbilical cord - erythema/induration
- Rule out Developmental Dysplasia of Hip (DDH).
- Breast feeding

Discharge check list

- Ensure discharge check list is completed. Discharge check list:
 - wt. gain / or no excess weight loss. Last weight should be mentioned in the discharge summary.
 - No ongoing problems
 - Feeding well established. Mother is comfortable with no feeding problems and latch is good.
 - Newborn screening (MS, hearing, eye, CCHD) is done. TSH screening is reviewed.(when applicable)
 - Immunization is done.
- Advise according to pamphlet & re-emphasize on
 - Exclusive breastfeeding till 6 months, complementary feeds from 6 months and breastfeeding along with complementary feeding till 2 years of age.
 - Sunlight exposure
 - No harmful cultural practices
 - Observing babies with adequate light
 - External feeds like water with herbs, butter, etc....
 - Follow up at 7 days of life.

Neonatal resuscitation

Identifying the high risk neonate

Antepartum Factors

Preterm delivery	Oligohydramnios
GA > 41 0/7 weeks	Fetal hydrops
Pre eclampsia / eclampsia	Fetal macrosomia
Maternal hypertension	IUGR
Multiple gestation	Fetal malformation
Fetal anemia	Maternal substance abuse
Polyhydramnios	No prenatal care

Intrapartum factors

Emergency cesarean section	Intrapartum bleeding
Forceps or vacuum-assisted delivery	Chorioamnionitis
Breech or other mal presentation delivery	Narcotics given < 4 h before
Category II or III FHR pattern	Shoulder dystocia
Maternal general anesthesia	MSAF
Maternal magnesium therapy	Proloapsed umbilical cord
Placental abruption	

Preparation for resuscitation

- Check inventory in the LR daily and enter in the inventory book

Anticipation and preparation are the key factors to effective resuscitation.

Enquire regarding gestational age, MSAF, singleton?, risk factors.

Personnel

- All deliveries to be attended by certified postgraduate with a nurse.
- < 32 weeks/ < 1500 g / high risk deliveries – senior resident or fellow
- < 28 weeks / CDH / < 1000 g – consultant when possible.
- Consider team briefing and role allocation

Equipment

Warmth

- Radiant warmer (heated for 10-20 min prior), 2 sterile warm towels, Cling wrap (GA < 32 wk)

Suction Equipment

- Mechanical slow suction (100 mm Hg) with tubing/bulb suction
- Suction catheters, 8F or 10F, and 12F or 14F

Positive pressure ventilation equipment

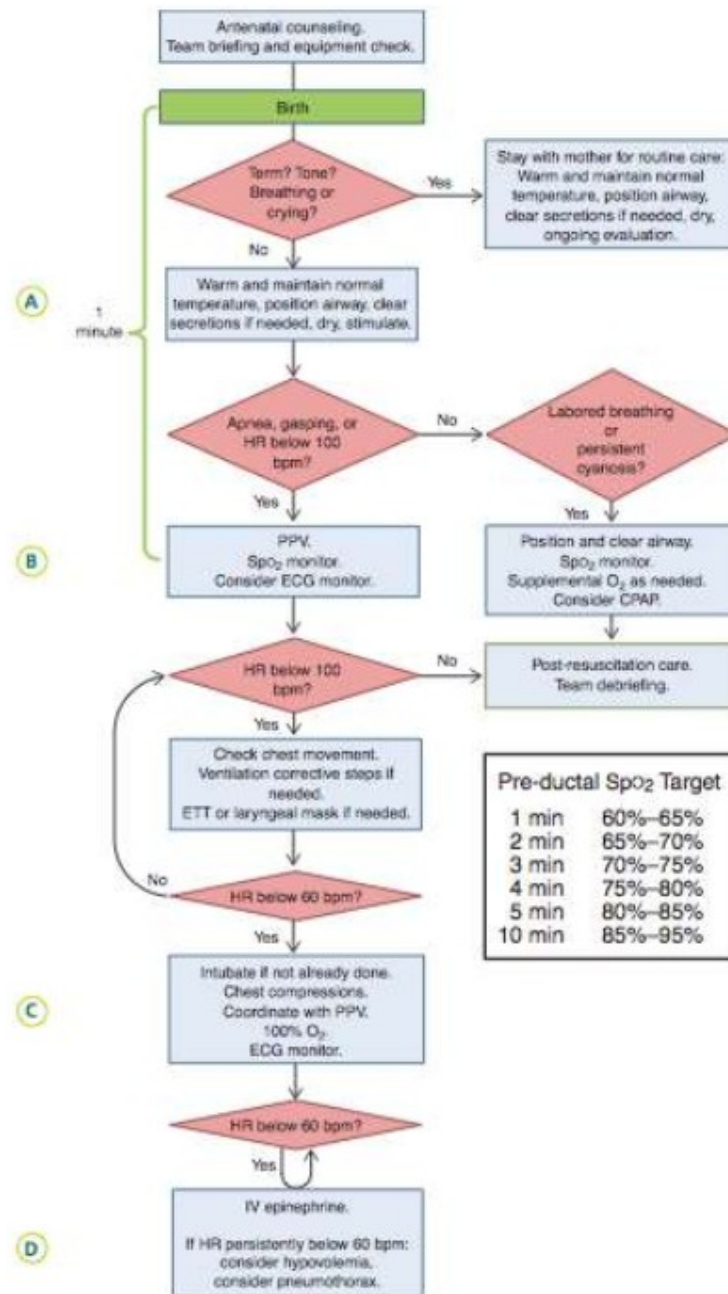
- Functioning self-inflating bag – with term and preterm mask
- T piece resuscitator (if preterm delivery)

Intubation equipment

- Laryngoscope with straight blades (No. 1 (term); 0 (preterm); 00 (extreme preterm))
- Endotracheal tubes: 2.5, 3.0, 3.5 (No. 2 tube not recommended)
- Tape for securing endotracheal tube

Umbilical vessel catheterization kit

- UVC kit – sterile cotton balls, gauze, sterile tie,
 - Sterile gloves, Scalpel or scissors, umbilical catheter (5 F feeding tube), 3 way stop cock, dynaplast, 1, 2, 10, 20 ml syringes
 - Spirit, Povidone - iodine solution, NS,
 - Daily freshly prepared 1:10000 adrenaline solution labeled with name, concentration and date of preparation. If used for a baby, rest is discarded and a fresh set is prepared for next use.
 - Others- Thermometer , stop watch
- * *Ensure that transport incubator is switched on and starting to heat while before going for delivery(especially for high risk delivery)*



Assess if the baby cries immediately after birth

If yes – provide Routine Care

- Provide routine care on the mothers abdomen
- If secretions are present, turn head to one side and wipe off secretions gently.
- Dry and discard wet cloth. Cover the head
- Cut cord after 1 minute in both term and preterm.
- Initiate breastfeeding as soon as baby is ready.
- Continue skin to skin care for at least 1 hour.
- All routine care such as Vit K, applying identification band, eye care can be done in skin to skin contact. Routines such as birth weight check, complete examination may be delayed. However if the baby is separated for these routines, the baby should be provided SSC as soon as possible.

Initial steps in resuscitation (30 seconds)

If baby does not cry immediately after birth, then the initial steps of resuscitation are provided. The initial steps in resuscitation consist of

- Preventing heat loss and providing warmth.
- Positioning and clearing the airway.
- Drying and removing the wet linen
- Initiating breathing by gentle stimulation (gently if required).
- Evaluation- HR, respiration and color

Evaluation of the newborn is based on

- Respiratory effort
- Heart rate

Positive pressure ventilation

Indications

- Apnea or gasping respiration
- Heart rate less than 100 per minute

Steps

1. Check the equipment to ensure that there are no tears or leaks.
2. Position the infant correctly with slight extension at the neck
3. Position the mask correctly and apply a firm seal. Provide 5 breaths and check for rising heart rate, if absent, adequate chest movement.
4. If there is no adequate chest rise, take ventilatory corrective steps (MRSOPA).
5. Once effective ventilation starts, continue for 30 seconds at 40-60 breaths / minute. Initiate PPV with room air (21% on blender; 21- 30% in preterm).

6. Connect to pulse oximeter. Titrate as per minute wise saturation targets. In preterm use 21% to 30% oxygen. If blender is not available, a self-inflating bag (with O₂ and without reservoir) provides 40%.
 - Pressures required to inflate the lungs :
 - Normal – 15 – 20 cm H₂O
 - Diseased – 20 – 40 cm H₂O
7. Evaluate the heart rate after 30 seconds of bag and mask ventilation
8. If heart rate is above 100 and infant has spontaneous respiration, discontinue ventilation
9. Heart rate between 60 & 100 – continue ventilation.
10. Heart rate < 60 beats per minute – ensure ventilation with 100% oxygen and initiate chest compression.
11. Pass an oro-gastric tube if bag and mask ventilation is continued for more than 2 minutes.

Contraindications

- Diaphragmatic hernia

Chest compressions

Indication

If heart rate less than 60/min despite good assisted ventilation for 30 sec.

Methods

- Two thumb method.

Steps

1. Intubate. Use 100% oxygen. Call for help if needed
2. Location: lower third of sternum strictly avoiding applying pressure on the xiphoid.
3. Depth: Compress the sternum approximately 1/3 of the A-P diameter of the chest.
4. Rate: 90 times per minute and ventilation 30 times per minute with a ratio of 3:1.
5. Evaluation: After 45-60 seconds of chest compression, check the heart rate.
6. If HR < 60/min- Continue compression and ventilation, Initiate medications.
7. If HR > 60/min - Discontinue compression. Continue ventilation till HR > 100/min
8. and good respiratory efforts.

Drugs

Epinephrine

Indications: HR < 60/min despite adequate ventilation and chest compression for 45-60 seconds. (at approximately 2 min of life) .

Dose: 0.1 – 0.3 ml/kg IV or 0.5 - 1 ml/kg Et 1 : 10,000 of epinephrine solution, may be repeated every 3-5 min if required. Endotracheal route is easier, but IV is preferred. Initiate UV cannulation as ET dose is given.

Volume expanders

- Normal saline, plasma expanders, whole blood can be used.
- Indication: Shock, acute bleeding, poor response to resuscitative efforts.
- Dose: 10 ml/kg IV over 5 –10 min.

Endotracheal intubation

Indication

- Bag and mask is ineffective or prolonged.
- Before chest compression
- Tracheal administration of medications is desired.
- Congenital diaphragmatic hernia
- ELBW for administering surfactant.

TUBE SIZE(ID mm)	B.WEIGHT(g)	GEST. AGEWks)
2.5	<1000	<28
3.0	1000-2000	28-34
3.5	2000-3000	35-38
4.0	>3000	>38

PLACENTAL TRANSFUSION

- Delayed cord clamping for all babies, both term and preterm who do not require resuscitation. Cord should be clamped at the end of 1 minute. This applies to both normal delivery and cesarean section.
- For babies requiring resuscitation, a 25 cm long segment of cord is secured with the help of cord clamping forceps. While initial steps are being done, the blood in the cord should be milked towards the baby.

Temperature Monitoring during Resuscitation:

- After the 60 seconds of PPV , consider switching to servo mode and monitor temperature.
- Check axillary temperature prior to shifting of infant.
- Pre-warm sheets prior to transfer of infant

Team Debriefing

The resuscitation team needs to discuss what went well and what did not go well and with points to focus after every resuscitation.

Necrotizing enterocolitis

- NEC is the most common life-threatening emergency of the gastrointestinal tract in the newborn period. characterized by various degrees of mucosal or transmural necrosis of the intestine.
- Cause of NEC
 - remains unclear
 - most likely multifactorial
 - incidence of NEC is 1-5% of infants in neonatal ICUs.
 - incidence and case fatality rates increase with decreasing birth weight and gestational age.
- Pathologic findings of NEC
 - necrotic segment of intestine
 - gas accumulation in the submucosa of the bowel wall (**pneumatosis intestinalis**)
 - progression of the necrosis to perforation,
 - peritonitis
 - sepsis
 - death

- The distal part of the ileum and the proximal segment of colon are involved most frequently; in fatal cases, gangrene may extend from the stomach to the rectum.
- Although NEC is a multifactorial disease primarily associated with intestinal immaturity, the concept of “risk factors” for NEC is controversial.
- **This triad has classically been linked to NEC.**
 - **intestinal ischemia (injury),**
 - **enteral nutrition (metabolic substrate), and**
 - **bacterial translocation**
- Prematurity is the greatest risk factor
- Rarely occurs before the initiation of enteral feeding and is much less common in infants fed human milk.
- Aggressive enteral feeding may predispose to the development of NEC.
- Coagulation necrosis is the characteristic histologic finding in intestinal specimens.

- Although nearly 90% of all cases of NEC occur in preterm infants, the disease can occur in full-term neonates.
- NEC in term infants is often a “secondary” disease,
 - more frequently in infants with history of
 - birth asphyxia
 - Down syndrome
 - congenital heart disease
 - rotavirus infections
 - Hirschsprung disease

History

- Age at onset
- Progression
- Preterm?
- Enteral feeding?
- lethargy and temperature instability, abdominal distention, gastric retention
- Type of stool, bloody stool
- Red cell transfusion
- History of
 - Birth asphyxia
 - Down syndrome
 - Congenital heart disease
 - Rotavirus infections
 - Hirschsprung disease

Table 102-1	Signs and Symptoms Associated with Necrotizing Enterocolitis
	GASTROINTESTINAL - Abdominal distention - Abdominal tenderness - Feeding intolerance - Delayed gastric emptying - Vomiting - Occult/gross blood in stool - Change in stool pattern/diarrhea - Abdominal mass - Erythema of abdominal wall
	SYSTEMIC - Lethargy - Apnea/respiratory distress - Temperature instability "Not right" - Acidosis (metabolic and/or respiratory) - Glucose instability - Poor perfusion/shock - Disseminated intravascular coagulopathy - Positive results of blood cultures

Diagnosis

- Very high index of suspicion in treating preterm at-risk infants
- Plain abdominal radiographs are essential to make a diagnosis of NEC
 - Pneumatosis intestinalis (air in the bowel wall) confirms the clinical suspicion of NEC and is diagnostic;
 - 50-75% of patients have pneumatosis when treatment is started
- Portal venous gas is a sign of severe disease, and pneumoperitoneum indicates a perforation
- Hepatic sonography may detect portal venous gas in some infants with normal abdominal x-rays.

Differential diagnosis

- Gastrointestinal obstruction
- Volvulus
- Isolated intestinal perforation
- Idiopathic focal intestinal perforation can occur spontaneously or after the early use of postnatal steroids and indomethacin.
 - Pneumoperitoneum develops in such patients, but they are usually less ill than those with NEC.

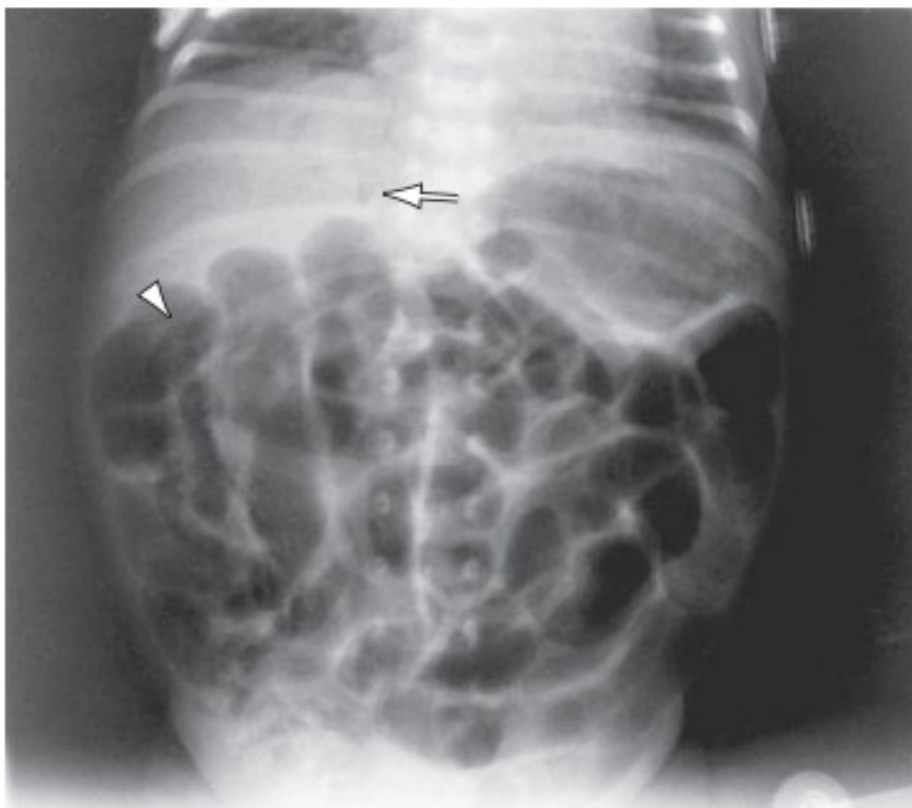


Figure 102-4 NEC. A kidney-ureter-bladder film demonstrates abdominal distention, hepatic portal venous gas (arrow), and a bubbly appearance of pneumatosis intestinalis (arrowhead; right lower quadrant). The latter 2 signs are thought to be pathognomonic for neonatal NEC.

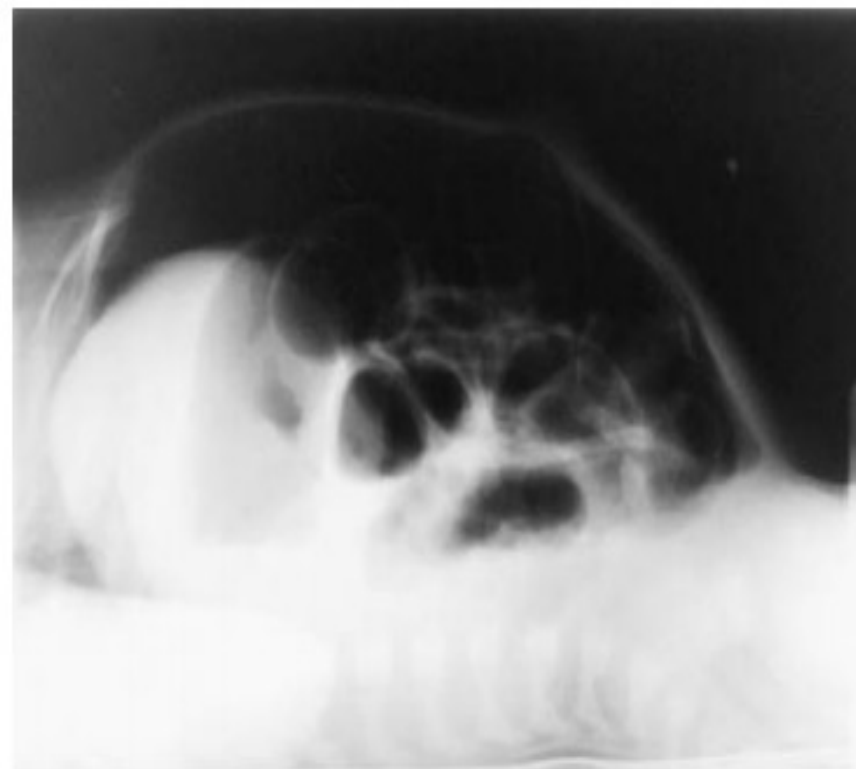


Figure 102-5 Intestinal perforation. A cross-table abdominal roentgenogram in a patient with a neonatal NEC demonstrates marked distention and massive pneumoperitoneum as evidenced by the free air below the anterior abdominal wall.

Treatment

- Rapid initiation of therapy
- No definitive treatment for established NEC
 - giving supportive care and preventing further injury with
 - cessation of feeding,
 - nasogastric decompression, and
 - administration of intravenous fluids.
- Broad spectrum antibiotics
- Intravascular volume replacement with crystalloid or blood products
- Surgery

Indications for surgery

- evidence of perforation on abdominal x-ray (pneumoperitoneum) or
- positive result of abdominal paracentesis (stool or organism on Gram stain preparation from peritoneal fluid)
- Failure of medical management,
- a single fixed bowel loop on radiographs,
- abdominal wall erythema, and
- a palpable mass are relative indications for exploratory laparotomy.

Prognosis

- Medical management fails in approximately 20-40% of patients with pneumatosis intestinalis at diagnosis; of these, 10-30% die.
- Early postoperative in approximately 10% of surgically or medically managed patients.
- Preterm infants with NEC who require surgical intervention or who have concomitant bacteremia are at increased risk for adverse growth and neurodevelopmental outcome.

Prevention

- Exclusive breast feeding
- Delayed or slow introduction of enteral feeding(controversial)
- Prophylactic antibiotics(Controversial)

TORCH

TORCH INFECTIONS

TORCH is an acronym

- Toxoplasmosis;
- Others.. syphilis, hepatitis B, coxsackievirus, Epstein-barr, varicella-zoster virus (VZV), and human parvovirus;
- Rubella virus;
- Cytomegalovirus [CMV] and
- Herpes simplex virus [HSV])

Modes of Transmission

	Transplacental	Perinatal
Toxoplasmosis	Yes	No
HBV	Yes	Yes
Syphilis	Yes	No
Rubella	Yes	No
CMV	Yes	Yes
HSV	Yes	Yes

TOXOPLASMOSIS

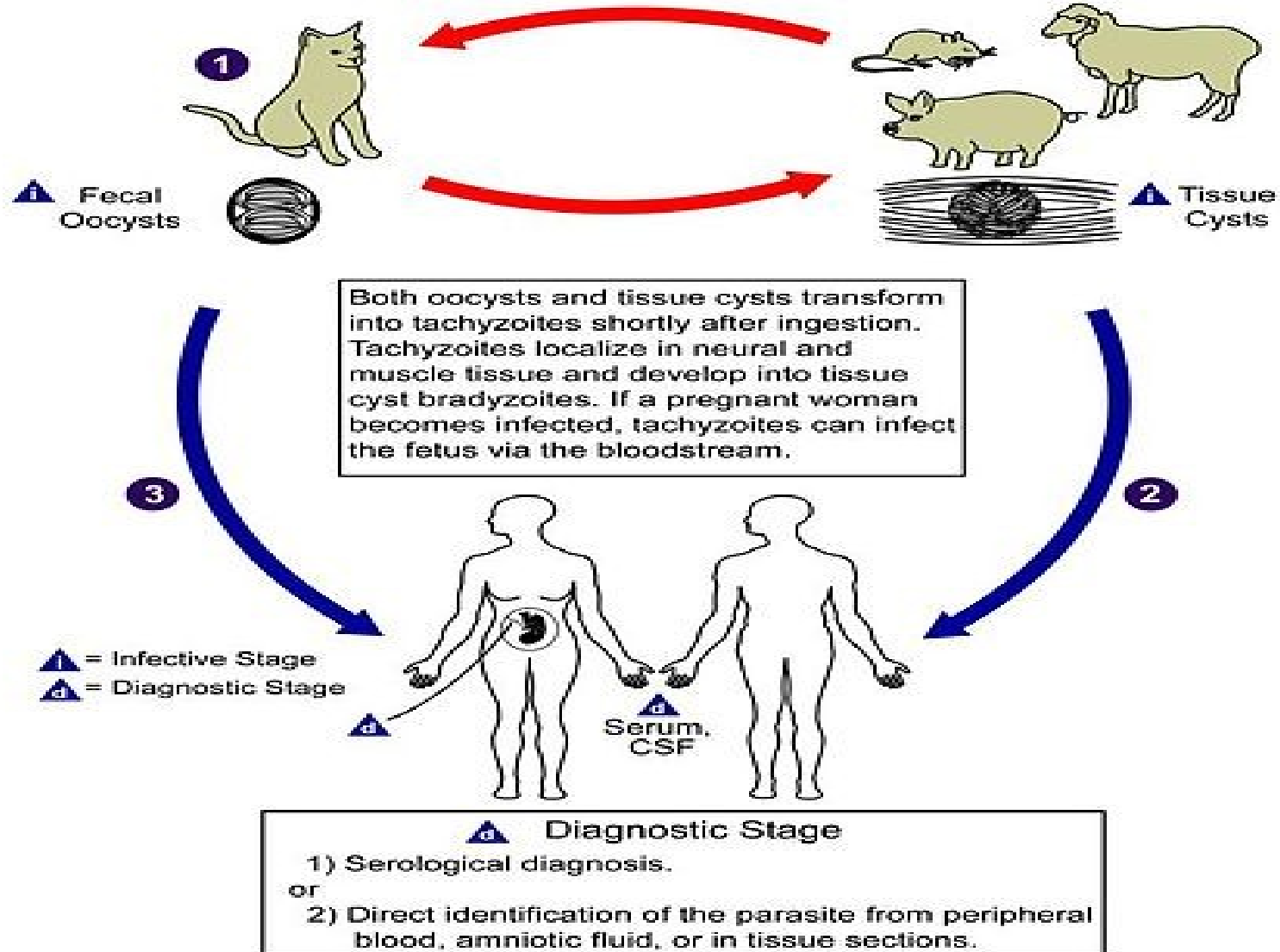
- Intracellular parasitic protozoan
- More risk as the pregnancy progresses 17% 1st TM and 65% 3rd TM
- The incidence of congenital infection is
 - 1 in 1000 to 1 in 10,000 live births

Risk factors

- *T. gondii* may be ingested during contact with soil or litter boxes contaminated with cat feces
- Unpasteurized milk, in raw or undercooked meats (especially pork)
- blood product transfusion (white blood cells)
- Prematurity

Toxoplasmosis

(*Toxoplasma gondii*)



Clinical presentation

- Congenital toxoplasmosis may be manifested as
 1. Clinical neonatal disease
 2. Disease in the first few months of life
 3. Late sequelae or relapsed infection or
 4. Subclinical disease
- Disseminated disease
- Isolated CNS or ocular disease
- Late sequelae are primarily related to ocular or CNS disease
- **The classic triad of toxoplasmosis**
 1. **Obstructive hydrocephalus**
 2. **Chorioretinitis, and**
 3. **intracranial calcifications form**

Diagnosis

- clinical suspicion
- Serologic tests:
 - IgG(The Sabin-Feldman dye test)
 - IgM
 - The IgG indirect fluorescent-antibody (IgG-IFA) test.
- PCR
- Direct isolation of the organism from body fluids or tissues
- Perinatal diagnosis
- CSF examination
- Radiologic studies
- Cranial ultrasonogram or CT scan of the head
- Long-bone films may show abnormalities
- Ophthalmologic examination

Treatment

- o Pyrimethamine
- o Sulfadiazine
- o Folinic Acid
- Spiramycine
- In case of signs of acute inflammation:
 - Corticosteroids (Prednisolone)

Complication

Blindness or severe visual disability

- Severe mental retardation
- Other neurological manifestations

Prevention

- Pregnant women should avoid
 1. eating raw meat or raw eggs and
 2. exposure to cat litter boxes or cat feces

RUBELLA

- Viral infection capable of causing
 - chronic intrauterine infection and
 - damage to the developing fetus
- Rubella vaccine has virtually eliminated cases of congenital rubella syndrome (CRS) in the developed countries
- The risk for congenital defects and disease is greatest with primary maternal infection during the first trimester.
- Risk factors
 - Women of childbearing age who are rubella nonimmune

Clinical presentation

- Wide spectrum of presentations
- Teratogenic effects
- Systemic involvement
- Later-presenting defects, normal at birth
- the majority later develop one or more signs and symptoms of disease

S

- Complications may involve many parts of the body.
- Triads of congenital rubella
 - Cataract
 - Patent ductus arteriosus
 - Microcephaly

Diagnosis

Confirmed by:

- finding rubella-specific IgM antibody in the neonatal serum, or
- culturing rubella virus from the infant (nasopharynx, urine, or tissues)
- Virus can be shed in the urine for 1 yr or longer

Management

- There is no specific treatment for rubella
- Long-term follow-up

CYTOMEGALOVIRUS

- CMV is a DNA virus member of the herpesvirus group
- Common cause of congenital infection
- Incidence is ~ 0.5-1.5% of all live births
- Most CMV infections are inapparent
 - but the virus can cause a variety of clinical illnesses that range in severity from mild to fatal
 - **CMV is the most common cause of congenital infection**

Risk factors

- lower socioeconomic status
- drug abuse
- sexual promiscuity in the mother
- Premature
- Transfusion with unscreened blood product

Clinical presentation

- Subclinical infection is 10 times more common than clinical illness
- LBW and SGA even when the infant is not infected
- IUGR
- Hepatosplenomegaly with jaundice
- Abnormal LFTs
- Thrombocytopenia with or without purpura
- Severe CNS disease in 50-90% of newborn
- Microcephaly
- Intracerebral calcifications (subependymal area),
- Chorioretinitis
- Progressive sensorineural hearing loss (10-20%)

Clinical presentation

- Hemolytic anemia
- Pneumonitis
- By 2 years of age, 5-15% of infants who are asymptomatic at birth may develop serious sequelae
 - such as hearing loss or
 - ocular abnormalities.
- Late sequelae
 - mental retardation
 - Learning disability and
 - sensorineural hearing loss in 7-15% bilateral
 - Progressive deterioration.

- cytomegalic inclusion disease

- Hepatosplenomegaly
- Jaundice
- Petechiae
- Purpura
- microcephaly

Diagnosis

- Active CMV infection is best confirmed by virus isolation from:
 - o Urine(positive within 48-72 hrs)
 - o Saliva(positive within 48-72 hrs)
 - o bronchoalveolar washings
 - o breast milk
 - o cervical secretions
 - o buffy coat or tissues obtained by biopsy.
- PCR

Treatment

– ganciclovir

SYPHILIS

- Sexually transmitted infection caused by the spirochete *Treponema pallidum*
- It can cross the placenta at any time during pregnancy
- The incubation period is typically about 3 weeks but could range from 9 to 90 days
- The disease has three clinically recognizable stages
 - Primary syphilis
 - Secondary syphilis
 - Tertiary syphilis

Risk factors

- lack of prenatal health care
- Maternal illicit drug use (cocaine use)
- No serologic test for syphilis (STS) performed during pregnancy

Clinical manifestation

- The early manifestations of congenital infection are varied and involve multiple organ systems.

Hepatosplenomegaly, jaundice, and elevated liver enzymes (common).

Lymphadenopathy

Coombs negative hemolytic anemia is characteristic.

Thrombocytopenia

osteocondritis and periostitis -

(pseudoparalysis of Parrot)

- ☐ mucocutaneous rash
- ☐ Mucous patches
- ☐ rhinitis (**snuffles**), and
- ☐ condylomatous lesions
- ☐ CNS abnormalities
- ☐ failure to thrive
- ☐ chorioretinitis
- ☐ nephritis, and nephrotic syndrome

- Clinical manifestations of renal involvement include:

- hypertension
- Hematuria
- Proteinuria
- Hypoproteinemia
- hypercholesterolemia, and
- hypocomplementemia.

- The late manifestations result primarily from chronic inflammation of:
 - bone
 - teeth, and
 - the CNS.
 - Skeletal changes due to persistent or recurrent periostitis and associated thickening of bone include:
 - frontal bossing
 - a bony prominence of the forehead (“**olympian brow**”)
 - unilateral or bilateral thickening of the sternoclavicular portion of the clavicle (**Higouménakis sign**)
 - an anterior bowing of the midportion of the tibia (**saber shins**)

- Dental abnormalities are common :
 1. **Hutchinson teeth** (peg or barrel-shaped upper central incisors)
 2. abnormal enamel, which results in a notch along the biting surface
 3. **mulberry molars**
 Defects in enamel formation lead to repeated caries and eventual tooth destruction.
 - A **saddle nose** (depression of the nasal root) is a result of syphilitic rhinitis that destroys the adjacent bone and cartilage.
 - **Rhagades** -are linear scars that extend in a spokelike pattern from previous mucocutaneous fissures of the mouth, anus, and genitalia

- **Juvenile paresis**, an uncommon latent meningovascular infection, typically presents during adolescence with:
 - o behavioral changes
 - o focal seizures, or
 - o loss of intellectual function.

Juvenile tabes with spinal cord involvement and cardiovascular involvement with aortitis are rare

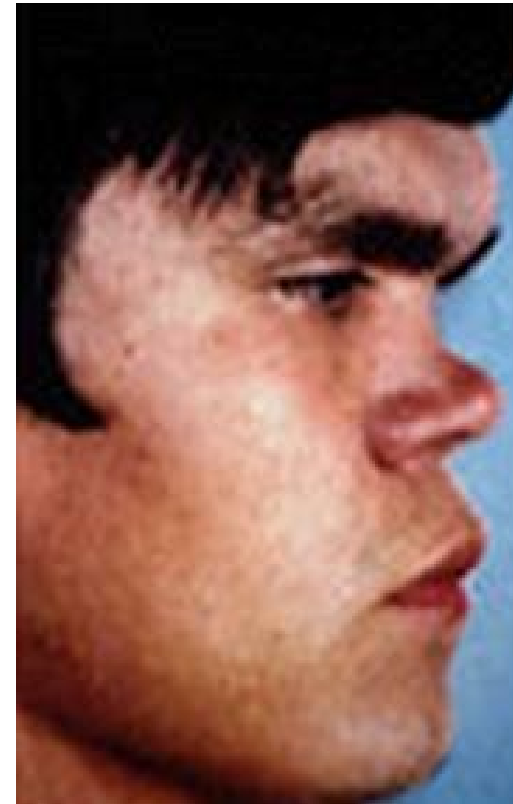
Hutchinson teeth



Mulberry molars



Saddle nose



Diagnosis

- Laboratory studies
 1. Nonspecific reagin antibody tests
 2. Specific treponemal tests
 3. Complete blood cell count with differential
 4. Lumbar puncture
 5. X-ray of the long bones

Management

- Treat the mother
- Treat the infant if maternal treatment was
 1. Inadequate
 2. Unknown
 3. Given during the last 4 weeks of pregnancy or
 4. Drug used was other than penicillin

Management

- For pregnant woman who has been treated for syphilis monthly quantitative NTA tests
- VDRL-positive infant treat with
- aqueous crystalline penicillin G 100,000-150,000 units/kg/24 h IV, or
- 50,000 units/kg/day of procaine penicillin IM for 10-14
- Aqueous crystalline penicillin G (100,000–150,000U/kg/24hr, given as 50,000U/kg IV BID for the first 7 days and TID thereafter) for 10–14 days OR
- Procaine penicillin G (50,000U/kg IM daily in a single dose) for 10–14 days

Isolation Procedures

- Precautions regarding drainage, secretions, blood and body fluids are indicated for all infants with suspected or proven congenital syphilis until therapy has been given for 24 h

HSV

- Neonatal HSV Infection
 - Most cases of neonatal herpes occur:
 - due to maternal infection during delivery
 - approximately 75% are HSV type 2.
 - Maternal primary or first episode genital herpes (no antibody present against the type of virus) has an infant attack rate of 33–50%.
 - Recurrent maternal disease has an infant attack rate of only 1–3%
- Perinatal infection manifests in the first month of life:
 - 9% on the first day, and
 - in 40% by the first week.
 - Three major categories:
 1. localized skin, eye, and mouth infection
 2. CNS infection and
 3. Disseminated infection.
 - Symptoms of CNS involvement are found in 48–79% and include:
 - o Lethargy
 - o poor feeding
 - o Irritability
 - o poor tone, and
 - o seizures

Diagnosis

- is based on any two of the following:

1. a compatible clinical pattern
2. isolation of the virus
3. development of specific antibodies and
4. demonstration of characteristic cells, histologic changes, viral antigen, or HSV DNA in scrapings, CSF, or biopsy material.

The use of PCR analysis of CSF and perhaps blood is critical to diagnose neonatal HSV infection, especially in the absence of skin lesions.

- The sensitivity of CSF PCR in neonatal CNS infection is 75%.
Viral culture is gold standard!
Microscopic examination of scrapings from lesions (Tzanck stain) reveals multinuclear giant cells and intranuclear inclusions approximately 50% of the time.

Treatment

Acyclovir

Hepatitis B

- Hepatitis B (serum hepatitis) is caused by a double-shelled DNA virus
- It has a long incubation period (45-160 days) after exposure

Risk factor

- Factors associated with higher rates of HBV transmission to neonates are:
 1. The presence of HBe antigen and absence of anti-HBe in maternal serum
 2. Maternal acute hepatitis in the third trimester or immediately postpartum (70% attack rate)
 3. Higher-titer HBsAg in maternal serum
 3. Antigenemia present in older siblings

Factors not related to transmission are

- A. The particular HBV subtype in the mother
- B. The presence or absence of HBsAg in amniotic fluid
- C. The presence or titer of anti-HBc in cord blood

Clinical presentation

- Maternal hepatitis B infection has not been associated with abortion, stillbirth, or congenital malformations
- Prematurity has occurred, especially with acute hepatitis during pregnancy. Fetuses or newborns exposed to HBV present a wide spectrum of disease
- Infants are usually asymptomatic
- jaundice appears in <3%
- Clinical presentations are various
 - Chronic active hepatitis with or without cirrhosis
 - Chronic persistent hepatitis
 - Chronic asymptomatic HBsAg carriage
 - Fulminant fatal hepatitis B (rare)

Diagnosis

Differential diagnosis

- biliary atresia
- acute hepatitis secondary to other viruses (eg, hepatitis A, CMV, rubella, and HSV)
- Mother Test for HBsAg, HBeAg, anti-HBe, and anti-HBc.
- Infant. Test for HBsAg and anti-HBc-IgM
- Transaminases
- Bilirubin (direct and indirect)
- Liver biopsy
- Hepatitis panel testing

Management

- Mother is HBsAg-positive, regardless of the status of her Hbe antigen or antibody, the infant should be given hepatitis B immune globulin (HBIG)
 - 0.5 mL IM, within 12 h after delivery
 - hepatitis B vaccine
- Recombivax (5 mcg [1.0 mL]) or Engerix (10 mcg [0.5 mL]) IM given at birth and at 1 month and 6 months of age.

Isolation

- Precautions are needed in handling blood and secretions
- Breast-feeding
- HBsAg has been detected in breast milk of HBsAg-positive mothers breast-feeding can be encouraged

Vaccine efficacy

- The overall protective efficiency rate in neonates given HBV vaccine and HBIG exceeds 93%.
- The HBV-infected neonate is usually asymptomatic but may develop mild clinical hepatitis and usually becomes a chronic carrier

Immunization program

- The WHO recommends that all countries add HBV vaccine to their routine childhood immunization program by 1997
- Two dose schedules have been proposed
- Each includes 3 separate doses
- In option 1, at birth, 1-2 months, and at 6-18months
- in option 2, at 1-2 months, 4 months, and 6-18 months.

Respiratory Distress in the Newborn

- Respiratory distress is a symptom complex arising from disease processes that causes failure to maintain adequate gaseous exchange and it will manifest as:-

- ✓ Cyanosis
- ✓ Grunting
- ✓ Retractions
- ✓ Tachypnea
- ✓ Nasal flaring
- ✓ Extreme: Apnea, Shock

Causes

- Respiratory distress syndrome (RDS)
- Pneumonia
- Meconium Aspiration
- Transient Tachypnea
- Hypothermia
- Hypoglycemia

Laboratory tests

- O₂ saturation
- X-ray: AP+lateral. Assess both lungs and heart
- Blood gas
- Hct
- Random blood sugar
- BP
- Transillumination
- Hyperoxia test
- Nasogastric catheter (radio opaque)
- Evaluate for sepsis

Management

- Clear airway, esp. meconium
- Oxygen
- Ventilation
 - mask bagging → intubation
 - Cyanosis
 - CO₂ retention
 - apnea
- Correct Acidosis

Respiratory Distress syndrome (Hyaline Membrane disease)

- Respiratory distress syndrome occurs primarily in premature infants
- Incidence is inversely proportional to gestational age and birth weight
- 60-80% in < 28 wk of gestational age
- 15-30% in between 32 and 36 wk
- 5% > 37 wk
- Rare at term
- Risk is higher in male infants

Risk factors

- Infants of diabetic mothers
- Delivery before 37 wk gestation
- Multifetal pregnancies
- Cesarean section delivery
- Precipitous delivery
- Asphyxia
- Cold stress
- History of previously affected infants

Decreased risk factors

- Chronic or pregnancy-associated hypertension
- Maternal opiate addiction
- Prolonged rupture of membranes
- Antenatal corticosteroid use

Surfactant deficiency

Surfactant:

- Phospholipids
 1. Main components:- phosphatidylcholine and phosphatidylglycerol
 2. Other constituents: cholesterol, neutral lipid and other phospholipids

Cont...

In the absence of surfactant:

- Progressive collapse of each alveoli during expiration
- Decreased compliance (more pressure needed to inflate same volume of air)
- Loss of functional residual capacity

Sequence in the premature infant:-

- Some surfactant present at birth
- Used up quickly
- Lungs unable to meet demand for more surfactant
- Baby develops progressive collapse of alveoli
- Hypoxia and acidosis further inhibit surfactant production, decrease pulmonary blood flow
- Hypoxia results in capillary damage and exudate into alveoli

Clinical presentation

- Premature infant
- Onset shortly after birth
- Grunting
- Retractions
- Nasal flaring
- Cyanosis
- Tachypnea
- Tachycardia
- Hypotension
- Reaches its peak on the third day
- Eventually...apnea

Differential diagnosis

- Group B streptococcal pneumonia
- Cyanotic congenital heart disease
- Persistent pulmonary hypertension
- Aspiration syndromes
- Congenital anomalies of the lung
- Air leak syndrome

Diagnosis

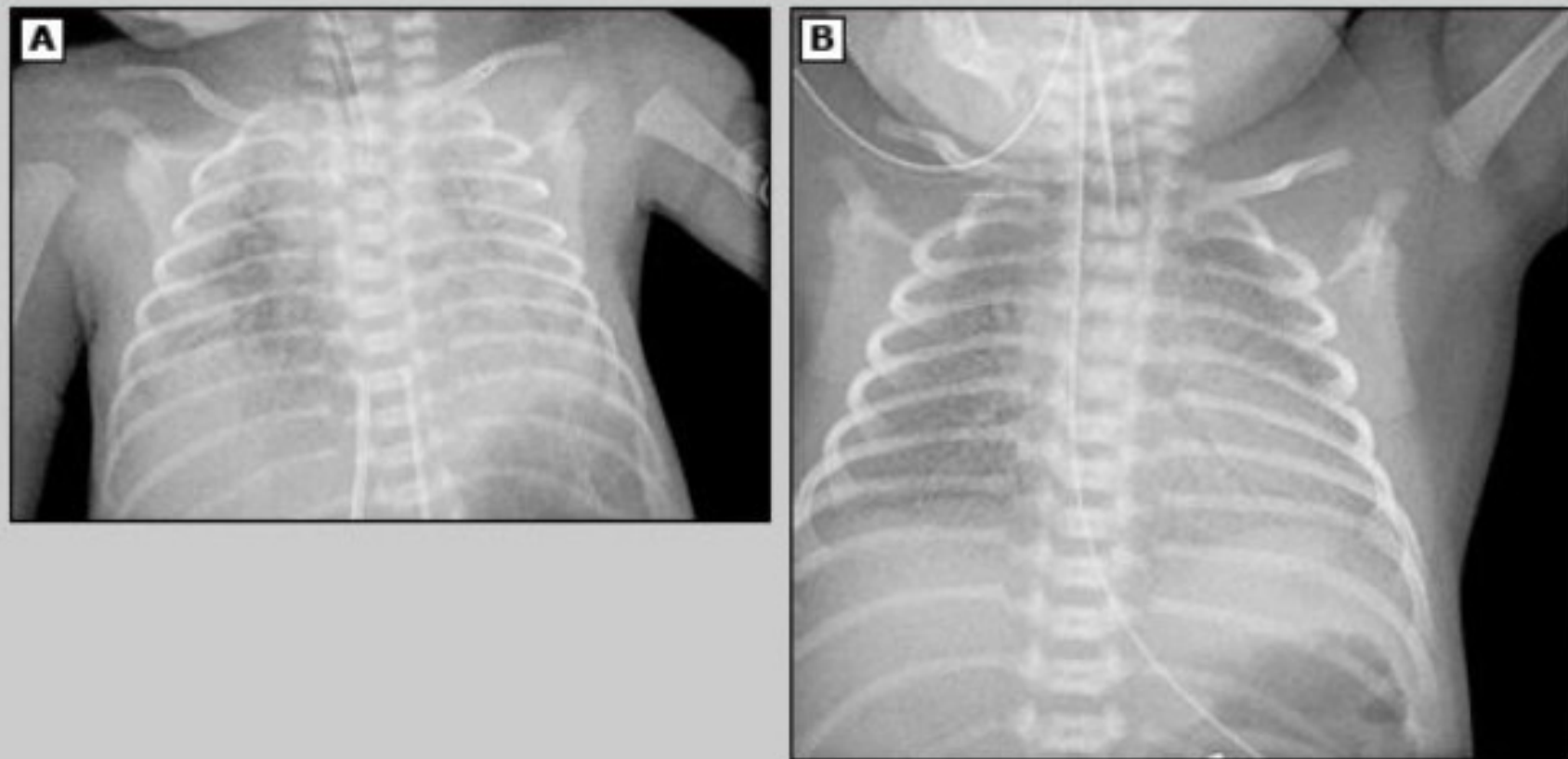
Blood gas features

- Acidosis
 - Metabolic (low HCO_3 , low Base Excess)
 - Respiratory (High pCO_2)
- Hypercarbia
 - Represents hypoventilation
- Hypoxia

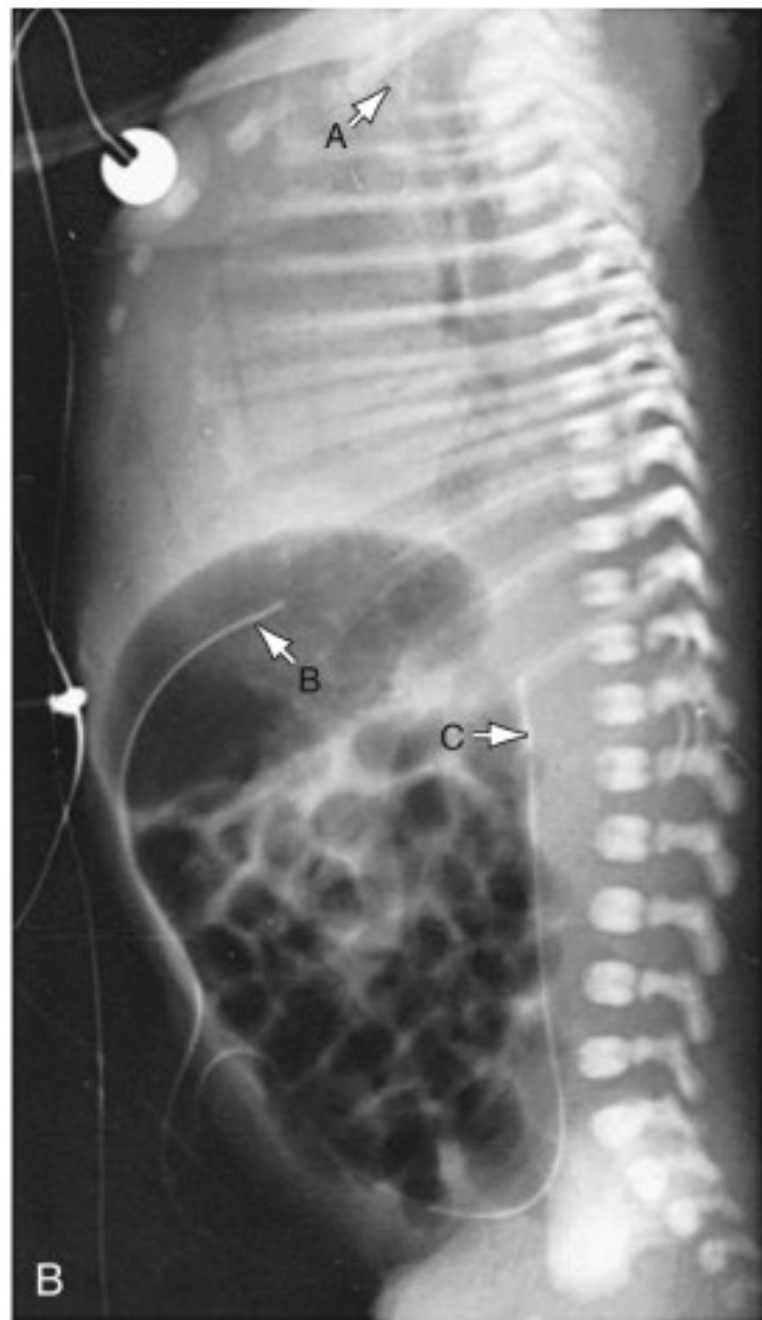
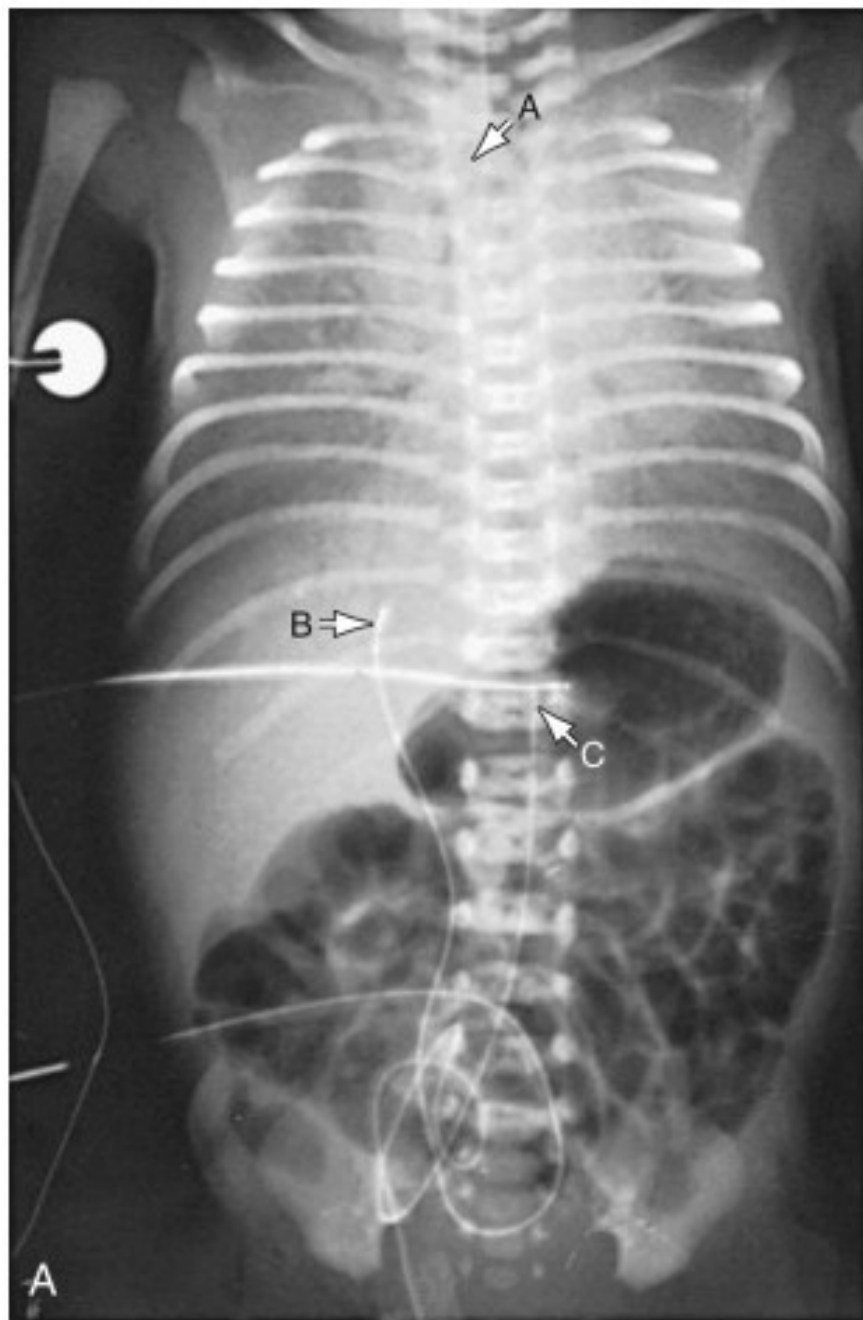
Radiographic features

- Poorly inflated lungs
- Hazy background, ground glass pattern
- Reticulo-granular pattern (clumps of collapsed alveoli)
- Air bronchograms

Neonatal respiratory distress syndrome



Two radiographs that demonstrate severe (A) and moderate (B) neonatal respiratory distress syndrome. Both demonstrate the characteristic low lung volumes, diffuse reticulogranular ground-glass appearance with air bronchograms.



Complications

- Air leak (pneumothorax, pneumomediastinum, pulmonary interstitial emphysema, pneumopericardium)
- Metabolic acidosis
- Hypotension
- Hypoxia
- Patent ductus arteriosus
- Chronic lung disease
- Other complications of prematurity (IVH, sepsis, NEC, ROP, etc)
- Bronchopulmonary dysplasia
- Patent ductus arteriosus
- Persistent fetal circulation
- Air leak syndrome related to CPAP
- Intra ventricular hemorrhage
- Necrotizing enterocolitis

Prevention

- Avoidance of unnecessary or poorly timed early cesarean section(< 39 wk) or induction of labor,
- Appropriate management of high-risk pregnancy and labor (including administration of antenatal corticosteroids),
- Prediction of pulmonary immaturity with possible in utero acceleration of maturation
- Prevention of fetal asphyxia
- Early CPAP at delivery room
- Prophylactic or early rescue surfactant

Treatment

- Oxygen
- Continuous positive airway pressure (CPAP)
- Mechanical ventilation with positive end expiratory pressure (PEEP)
- Surfactant (endotracheal administration)
- In severe cases, nitric oxide, high frequency ventilation
- Aims of therapy:
 - pH >7.25
 - pO₂ 50-70 mmHg
- pCO₂ 50-60 mmHg (moderateEarly supportive treatment of hypoxia, acidosis, hypotension and hypothermia
- Cause we can't rule out infection initiation of antibiotics is warranted
 - hypercapnea)

Complications of prematurity

Acute

- Respiratory distress syndrome (HMD)
- Hypothermia
- Hypoglycemia
- Sepsis
- Necrotizing enterocolitis
- Intraventricular hemorrhage
- Retinopathy of prematurity
- Pathologic jaundice

Late onset

- Anemia of prematurity
- Retinopathy of prematurity
- Bronchopulmonary dysplasia
- Developmental delay
- Mental retardation
- Hearing and vision deficit
- Patent ductus arteriosus

Management

1. Manage HMD using CPAP, surfactant
2. Treat hypoglycemia via 10% d/w then early priming of breast milk
3. Incubators to control body temperature
4. Follow for other complications like ROM, Jaundice, NEC, IVH and PDA
5. Advise mother to start KMC
6. Breast and preterm formula for optimal growth

Follow up

- Follow weight gain at least 15-20gm/kg/day
- Follow for head circumference 1cm/wk
- Anthropometric measurement using corrected age
- Vision assessment
- Hearing assessment
- Brain ultrasound
- Development assessment
- Assess for anemia, rickets and bleeding tendencies

Complications of post term

Fetal macrosomia

Feeding practices

- **Infant and young child feeding:**

- **Exclusive breast feeding:**

- the baby is given only breast milk and nothing else.
 - Should be practiced during the first six month.
 - Breast milk provides best and complete nourishment to the baby during the first six months.

Ingredient	amount
Calorie	67kcal/dl
Protein	1.8-3.1gm/dl
whey	70%
casein	30%
nucleotides	Present
Lipid	50% of the calorie in the milk
vitamins	
minerals	

Feeding...

- **Advantage of breast milk:**

- Improved gastrointestinal function, digestion, and absorption
- Improved cognitive and visual development
- Improved host defense
- Enhanced maternal psychological well-being and maternal-infant bonding.
- Increase the rate of gastric emptying .
- Increase the intestinal lactase activity in premature infants.
- Decrease the intestinal permeability early in life in premature infants .
- Decrease the risk of NEC in premature infants.

- **Advantage of breast milk(continued)**

- Acceleration of uterine involution.
- Reduction of maternal response to stress
- Lower rates of maternal neglect and child abuse
- Enhanced weight loss after pregnancy.
- Prolongation of postpartum anovulation.
- Economic benefit.

Feeding...

- **Signs of good attachment in breast feeding:**

- Mouth wide open
- The lower lip is turned outward
- The chin is touching the breast
- More areola above than below

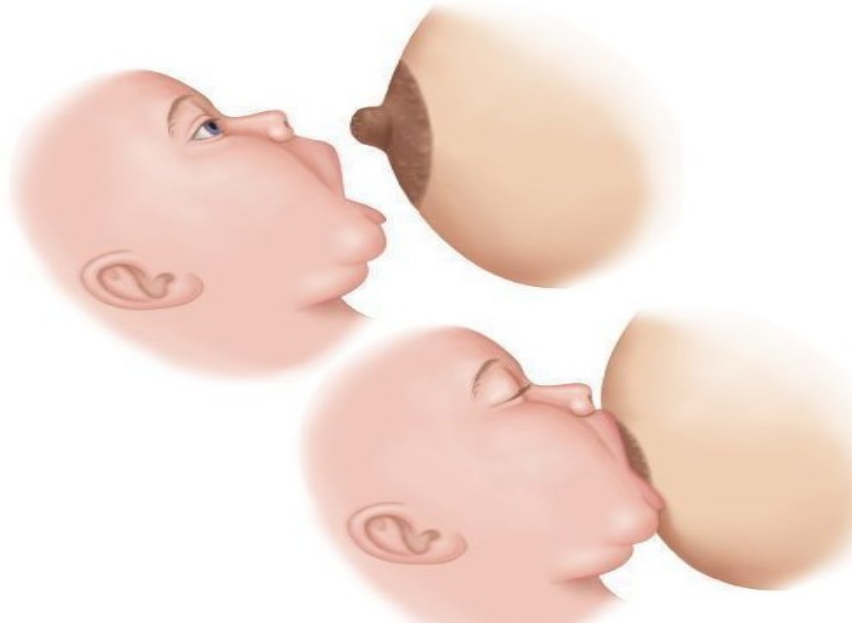
- **Positions when breast feeding:**

- The cradle hold
- The cross cradle hold
- The seated position

- **Signs of good positioning:**

- The baby's stomach should touch the mother's abdomen.
- the baby's body should be straight.
- The baby's body has to be supported.

Infant's latch-on for breastfeeding



Cross-cradle position for breastfeeding



Cradle position for breastfeeding



Seated nursing position for breastfeeding



Feeding...

- **Assessment of breast milk intake adequacy:**

- nurses 8 to 12 times in 24 hours
- Infant urinate six to eight times daily
- Passes three or more pale yellow and seedy stools daily.
- Will have adequate weight gain.

- **Complementary feeding:**

- **Importance of complementary feeding:**

- Is essential from six months of age while continuing breast feeding.
- The purpose is to complement breast milk to provide enough **energy, protein** and **other nutrients**.
- Breast feeding has to be continued up to the **age of two years** or beyond.
- Active feeding style for complementary feeding is important.

Feeding...

- **First food for the baby:**

- Cereal food is recommended
- Porridge, milk, ground rice, fitfit, smashed potato, smashed vegetables, fruits...
- Adding ghee or oil is good as it increases the energy value of the food.
- Gradually egg, meat and fish can be introduced.

- **Energy density of food can be increased by:**

- By adding ghee or oil to the food
- By adding sugar to the food
- By feeding thick mixtures

Feeding...

- **Frequency of feeding:**

- Infants and young children need to be fed 5-6 times a day in addition to breast feeding.
- Inadequate feeding is the most common cause of malnutrition.

- **Feeding during and after illness:**

- One or two extra meals has to be given during this time.

Neonatal hyperbilirubinemia

Etiology

- Unconjugated hyperbilirubinemia may be caused by:
 1. increased load of bilirubin to be metabolized by the liver
 2. damage or reduced activity of the transferase enzyme
 3. competing or blocking the transferase enzyme (drugs)
 4. absence or decreased amounts of the enzyme (genetic defect, and prematurity).

Clinical feature

- apparent in a cephalocaudal progression:
 - jaundice of face ≈ 5 mg/dL
 - mid-abdomen ≈ 15 mg/dL
 - soles, ≈ 20 mg/dL
- Infants may present with **lethargy and poor feeding**.
- without treatment, can progress to **acute bilirubin encephalopathy (kernicterus)**.

Classification of jaundice

Physiologic jaundice	Pathologic jaundice
Jaundice appearing after 24hr	Appears in the first 24hrs of life
Rate of rise is $<5\text{mg/dl/24hr}$	$>5\text{mg/dl/24hr}$ rate of rise
TSB $<12\text{mg/dl}$ for term and $<15\text{mg/dl}$ for preterm	TSB is $>12\text{mg/dl}$ for term and $>12\text{mg/dl}$ for preterm
Jaundice doesn't persist more than 2weeks	Jaundice persists after 2weeks
Direct bilirubin is $<2\text{mg/dl}$	Direct bilirubin is $>2\text{mg/dl}$

Differential diagnosis

- Jaundice with in the first week:
 - erythroblastosis fetalis
 - concealed hemorrhage
 - Sepsis
 - congenital infections
 - Crigler Najjar syndrome
 - polycythemia
- *after* the 1st wk of life:
 - breast milk jaundice
 - Septicemia
 - congenital atresia or paucity of the bile ducts
 - Hepatitis
 - Galactosemia
 - Hypothyroidism
 - RBC morphology and enzyme deficiencies

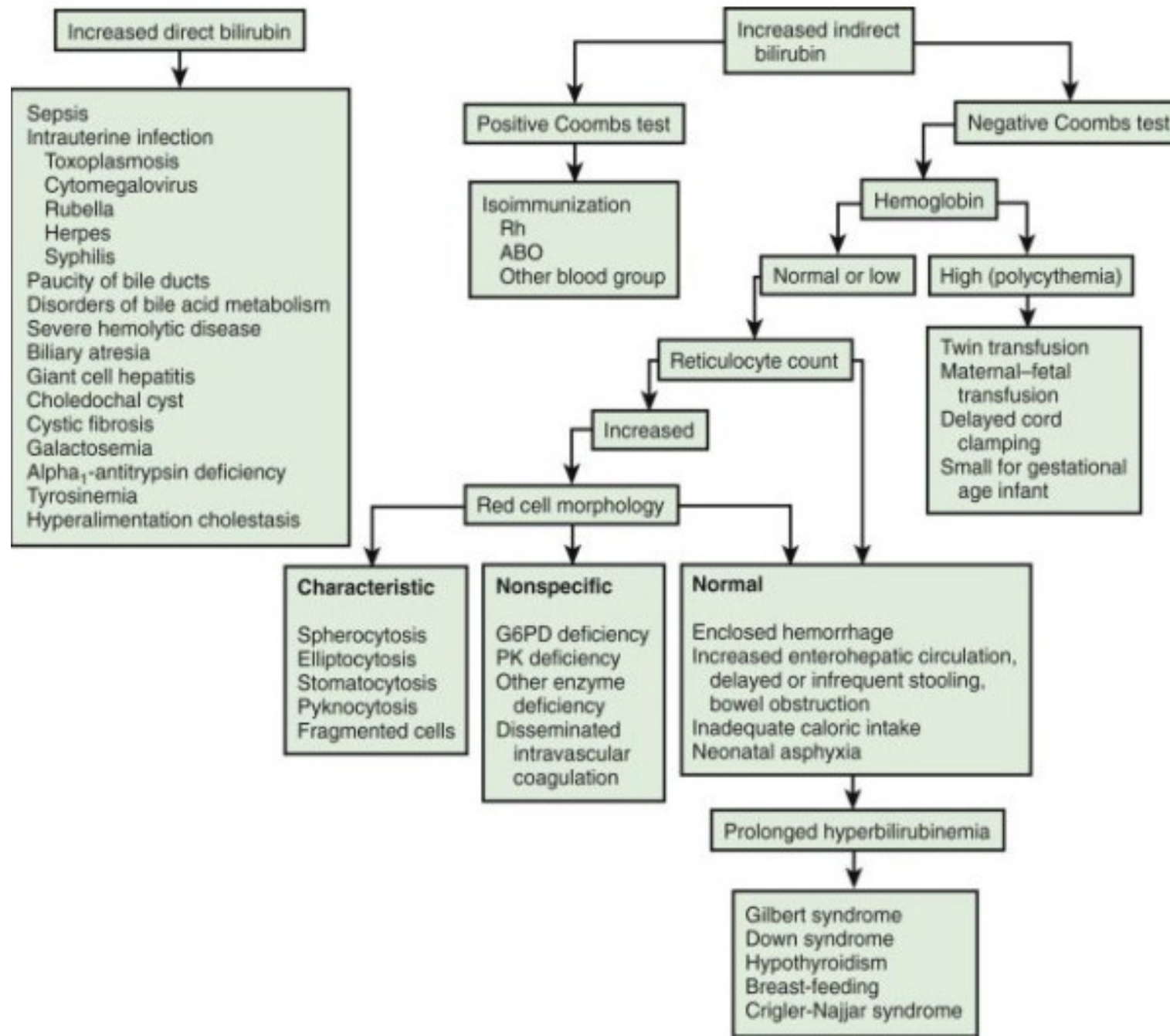
Differential diagnosis...

- persistent jaundice during the first month of life:
 - Hepatitis
 - congenital infections
 - congenital atresia of the bile ducts
 - Galactosemia (genetic disease (autosomal recessive) in which an enzyme needed to metabolize galactose is deficient or absent; typically develops shortly after birth)
 - inspissated bile syndrome following hemolytic disease of the newborn.

Investigations

- direct and total bilirubin
- Hemoglobin/hematocrite
- reticulocyte count
- blood type
- Coombs test
- peripheral blood smear
- If direct hyperbilirubinemia, investigate for underlying factors

Differential diagnosis...



Bilirubin associated with breast feeding

- **Breast milk jaundice:**

- develops in an estimated 2% of breast-fed term infants after the 7th day of life.
- If nursing is discontinued, the serum bilirubin level falls rapidly.
- may be attributed to the presence of glucuronidase in some breast milk.

- **Breast feeding jaundice:**

- occurs in the 1st week of life.
- may be due to decreased milk intake with dehydration.
- Frequent breast-feeding is the treatment.

Hemolytic disease of the newborn (erythroblastosis fetalis)

- caused by the **transplacental passage of maternal antibody active against paternal RBC antigens of the infant.**

● Pathogenesis:

- When **Rh-positive blood is infused into an Rh-negative woman**, antibody formation against D antigen may be induced in the unsensitized Rh-negative recipient mother.
- **IgG antibody readily crosses the placenta to cause hemolytic manifestations.**
- Hemolytic disease **rarely occurs during a first pregnancy because transfusion of Rh-positive fetal blood into an Rh-negative mother occurs near the time of delivery.**
- The severity of Rh illness worsens with successive pregnancies.

Rh incompatibility...

- **Clinical features:**

- Severity ranges from mild hemolysis to severe anemia.
- Jaundice on the first day of life.
- Enlargement of liver and spleen.
- anemia
- signs of cardiac decompensation
- anasarca
- **hydrops fetalis-** excessive abnormal fluid in two or more fetal compartments.

Investigatin

- direct Coombs test is usually positive
- Anemia
- blood smear - polychromasia and reticulocyte count is increased.
- Thrombocytopenia- in severe cases
- Cord bilirubin is generally between 3 and 5 mg/dL

Rh incompatibility...

- **Antenatal diagnosis:**

- Maternal sensitization
- Doppler ultrasonography of the middle cerebral artery
- percutaneous umbilical blood sampling (PUBS)
- U/S signs of hydrops

- Unborn infant- transfusion

- Live born infant:

- ✓ If severe hemolytic anemia- immediate resuscitation and supportive therapy, temperature stabilization, and monitoring.
- ✓ exchange transfusion
- ✓ Phototherapy
- ✓ IVIG

Rh incompatibility...

- **Prevention of Rh sensitization:**

- Sensitization reduced to less than 1% by the IM injection of 300 ug of human anti-D globulin.
- anti-D administration at 28-32 wk and again at birth (40 wk) is more effective than a single dose.

ABO incompatibility

- is the most common cause of hemolytic disease.
 - results in milder disease than Rh incompatibility does.
 - Low antigenicity of the ABO factors in the fetus and newborn infant may account for the low incidence of severe ABO hemolytic disease.
 - the mother is type O and the infant is type A or B.
- **Clinical feature:**
 - Most cases are mild
 - infant is not generally affected at birth
 - hydrops fetalis is extremely rare
 - Rarely, it may become severe, and symptoms and signs of kernicterus develop rapidly.
 - **Treatment:**
 - Phototherapy
 - IVIG
 - Exchange transfusion

Complications associated with phototherapy

- **loose stools**, erythematous macular rash, purpuric rash
- overheating, dehydration, hypothermia from exposure, and
- bronze baby syndrome
- Before phototherapy is initiated, the infant's eyes should be covered
- Body temperature should be monitored

Potential complications from exchange transfusion

- metabolic acidosis, electrolyte abnormalities, hypoglycemia, hypocalcemia,
- thrombocytopenia, volume overload, arrhythmias,
- NEC, infection, graft-versus-host disease, and death.

Hemorrhagic disease of the new born

- Hemorrhagic Diseases of the Newborn/HDN

- Temporary coagulation defect due to vitamin K deficiency

- causing spontaneous hemorrhage

- Pathophysiology of HDN

- o Deficiency of Vitamin K-dependent factors (II, VII, IX, X)

- o Common during the 1st 48-72hrs

- o Vitamin K deficiency occurs due to:

- Poor source from the mother (placental, Breast milk)

- Absence of bacterial flora in the newborn's intestine

- Immaturity of newborn's liver

- Investigations

- Prothrombin Time(PT)--prolonged

- aPTT--prolonged

- Clotting time—prolonged

- Reduced vitamin k –dependent factors

- Normal BT, Platelet and fibrinogen

- Treatment

- o Vitamin K 1mg Im at birth (prophylaxis)

- o Vitamin K 1-5mg (IV slow infusion)—therapeutic

- o Plasma transfusion (severe cases)

- Complications

- Intracranial Bleeding (serious)

Table 103-5	Hemorrhagic Disease of the Newborn		
	EARLY-ONSET DISEASE	CLASSIC DISEASE	LATE-ONSET DISEASE
Age	0-24 hr	2-7 days	1-6 mo
Site of hemorrhage	Cephalohematoma Subgaleal Intracranial Gastrointestinal Umbilicus Intraabdominal	Gastrointestinal Ear-nose-throat-mucosal Intracranial Circumcision Cutaneous Gastrointestinal Injection sites	Intracranial Gastrointestinal Cutaneous Ear-nose-throat-mucosal Injection sites Thoracic
Etiology/risks	Maternal drugs (phenobarbital, phenytoin, warfarin, rifampin, isoniazid) that interfere with vitamin K Inherited coagulopathy	Vitamin K deficiency Breastfeeding	Cholestasis—malabsorption of vitamin K (biliary atresia, cystic fibrosis, hepatitis) Abetalipoprotein deficiency Idiopathic in Asian breastfed infants Warfarin ingestion
Prevention	Possibly, administrations of vitamin K to infant at birth or to mother (20 mg) before birth Avoid high-risk medications	Prevented by parenteral vitamin K at birth Oral vitamin K regimens require repeated dosing over time	Prevented by parenteral and high-dose oral vitamin K during periods of malabsorption or cholestasis
Incidence	Very rare	≈ 2% if infant not given vitamin K	Dependent on primary disease

Neonatal Seizures

TYPES OF NEONATAL SEIZURES

- There are **5 main neonatal seizure** types:
 - subtle
 - clonic
 - tonic
 - spasms
 - myoclonic.
- As a rule seizures, associated with electrographic discharges (epileptic seizures) are,
 - **Spasms**
 - **focal clonic**
 - **focal tonic**
 - **generalized myoclonic**

Subtle Seizures

- Subtle seizures include
 - ❖ transient eye deviations,
 - ❖ nystagmus
 - ❖ blinking
 - ❖ mouthing,
 - ❖ abnormal extremity movements (rowing, swimming, bicycling, pedaling, and stepping),
 - ❖ fluctuations in heart rate,
 - ❖ hypertension episodes, and
 - ❖ apnea.
- Subtle seizures occur more commonly in premature than in full-term infants.

Seizures vs Jitteriness

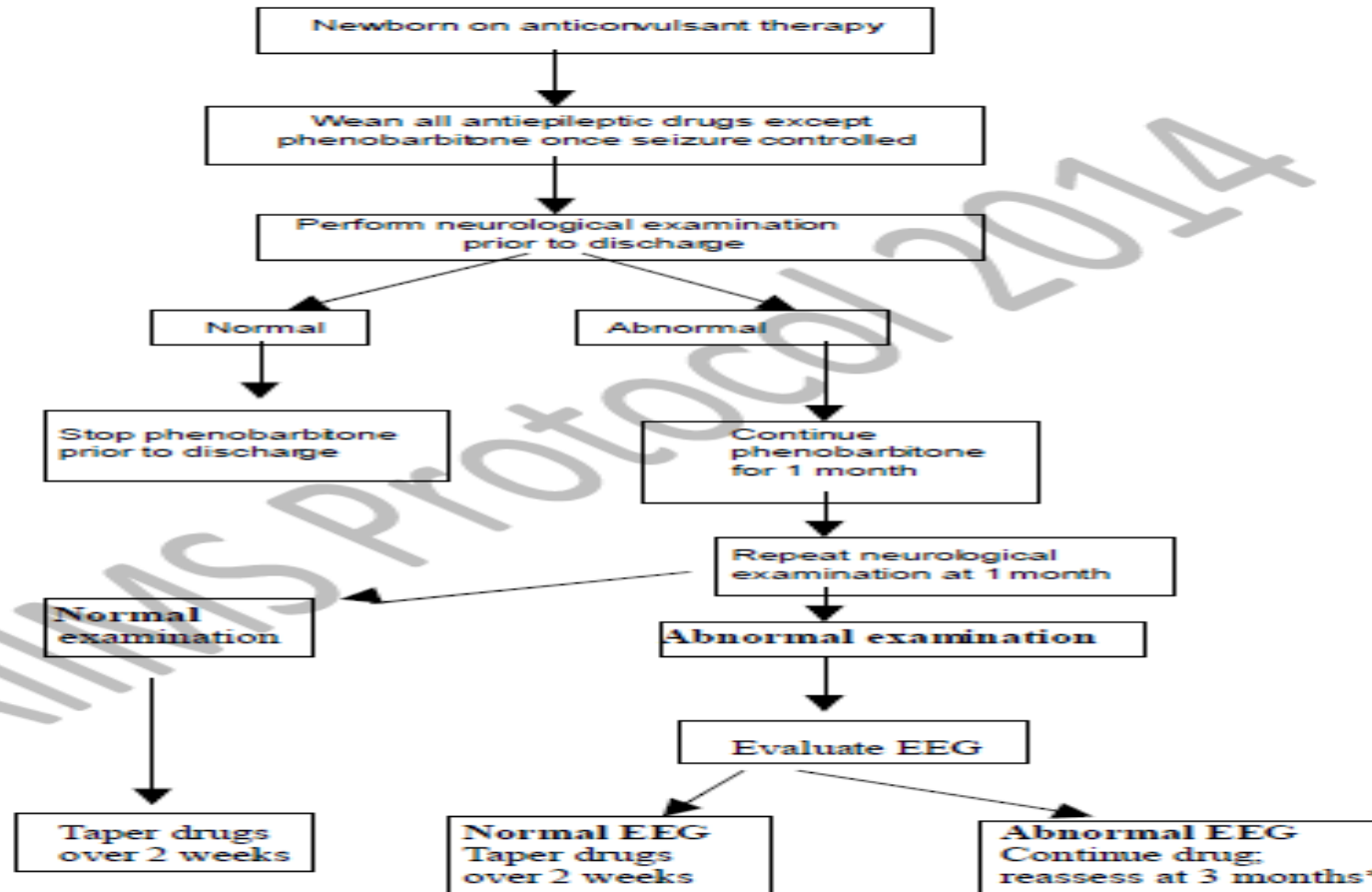
- Jitteriness can be defined as rapid motor activities, such as a tremor or shake
- can be ended by flexion or holding the limb.
- Seizures, on the other hand, generally do not end with **tactile or motor** suppression.
- Jitteriness, unlike most seizures, is usually **induced by a stimulus**.
- Also unlike jitteriness, seizures often **involve eye deviation and autonomic changes**

Treatment

- A mainstay in the therapy of neonatal seizures is
 - ☞ the diagnosis
 - ☞ treatment of the underlying etiology (e.g., hypoglycemia, hypocalcemia, meningitis, drug withdrawal, trauma), whenever one can be identified
- Phenobarbital is considered by many as the first choice long-acting drug in neonatal seizures.
- The usual loading dose is 20 mg/kg.
- For ongoing seizures, if a total loading dose of 40 mg/kg of phenobarbital was not effective,
- then a loading dose of 15-20 mg/kg of phenytoin can be administered intravenously.

Duration of Therapy

Figure 2 Weaning of anticonvulsant therapy



**Intractable seizures may need lifelong therapy; consider switching over to other drugs (phenytoin or carbamazepine)*

Management

- The goal is **to stop electrographic seizure** activity before reducing the therapy
- Principles of management
 1. Treat life threatening conditions
 2. Treat the acute seizure
 3. Treat the underlying cause
 4. Prevent recurrence
 5. Prevent and treat systemic complications

NEONATAL SEIZURE

- Record the type and duration of convulsions
 - **Subtle:** horizontal deviation of eyes, staring, fluttering, chewing, tongue, thrusting, lip smacking, cycling, peddling, rowing, apnea.
 - **Clonic:** focal or multifocal.
 - **Tonic :** focal or generalized Myoclonic : focal, multifocal or generalized
- Nurse the baby in thermo- neutral environment and ensure airway, breathing and circulation.
- Confirm hemodynamic stability of the infant.
- If stable, secure an IV line
- If unstable, take appropriate resuscitative measures.
- Perform RBS with glucometer.
- If less than 40 mg/dl, administer 10% dextrose, 2 ml/kg followed by glucose infusion drip at a rate of 6-8 mg/kg/min. For further
- Management of hypoglycemia (see hypoglycemia protocol)
- If RBS > 40 mg/dl and if convulsions continue then give anticonvulsants Once convulsions stops, collect blood for serum electrolytes , RBS, calcium, phosphorus and alkaline phosphatase, magnesium.

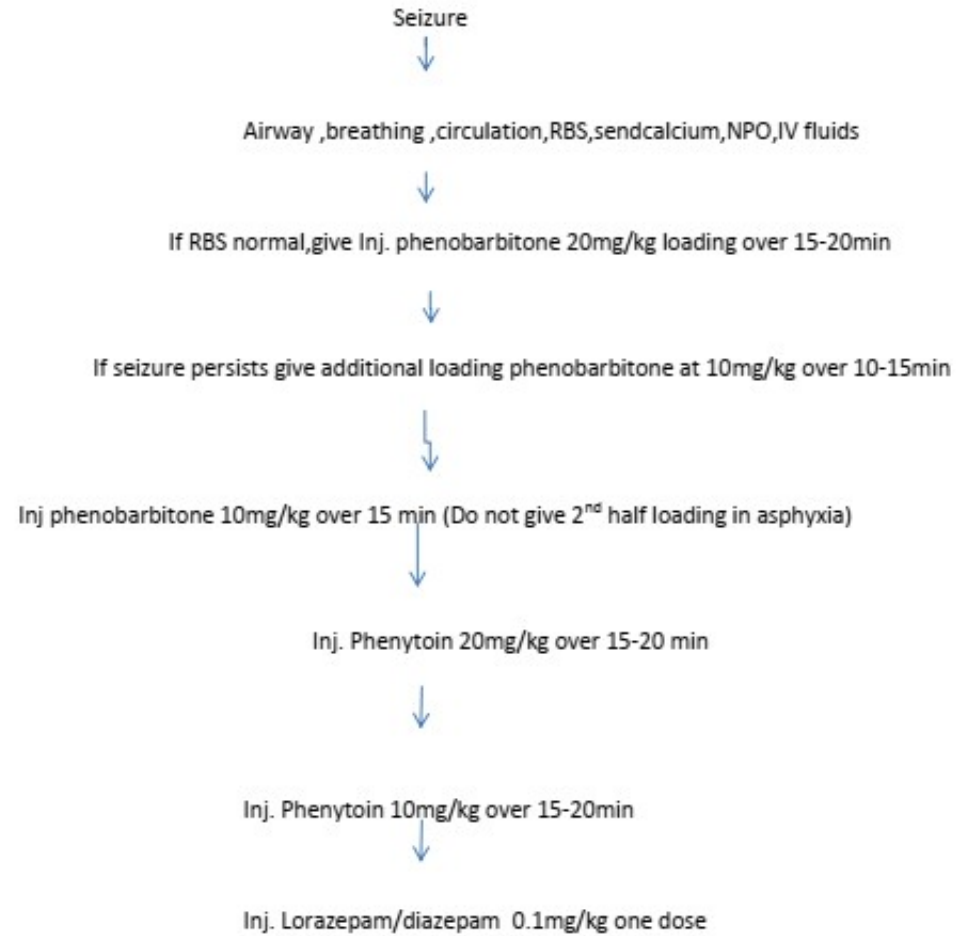
Etiology:

- Asphyxia
- Intracranial bleeds
- Metabolic – hypoglycemia, hypocalcaemia, hypomagnesaemia, electrolyte disturbances, IEM
- Infection – Meningitis
- Cranial malformations
- Miscellaneous – Benign sleep myoclonus, Fifth day seizures

Investigations

- RBS , calcium, magnesium, serum electrolytes
- Septic profile
- Cranial USG, MRI / CT Scan if indicated
- aEEG/EEG

ALGORITHM FOR MANAGEMENT OF NEONATAL SEIZURE



Cleft lip and palate

Cleft Lip and Palate

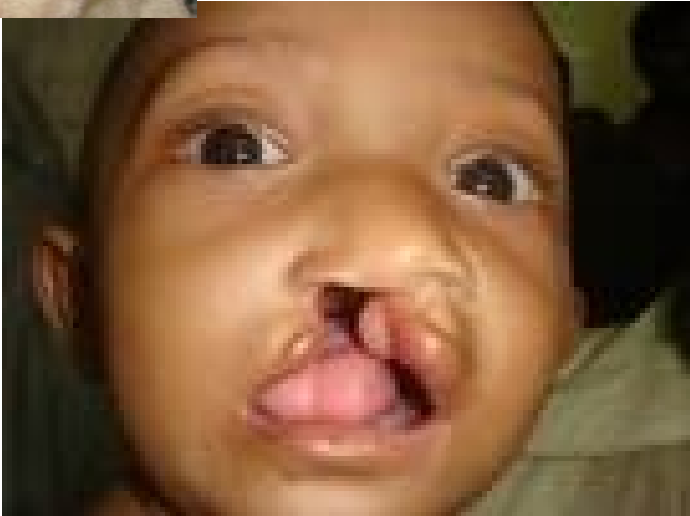
- Clefts of the lip and palate are distinct entities closely related embryologically, functionally, and genetically.
- A cleft lip or palate is an abnormal separation in the oral-facial region that happens because tissue of the mouth or lip does not form correctly in fetal development.
- resulting in a failure of the medial nasal and maxillary processes to join.
- Cleft of the palate appears to represent failure of the palatal shelves to approximate or fuse
- The incidence of cleft lip with or without cleft palate is $\approx 1/750$ white births;
- the incidence of cleft palate alone is $\approx 1/2,500$ white births.
- Clefts of the lip are more common in males.
- Possible causes include
 - maternal drug exposure,
 - a syndrome-malformation complex, or
 - genetic factors.
 - the recurrence risk is 50%.

Cleft Lip and Palate...



09/15/2026

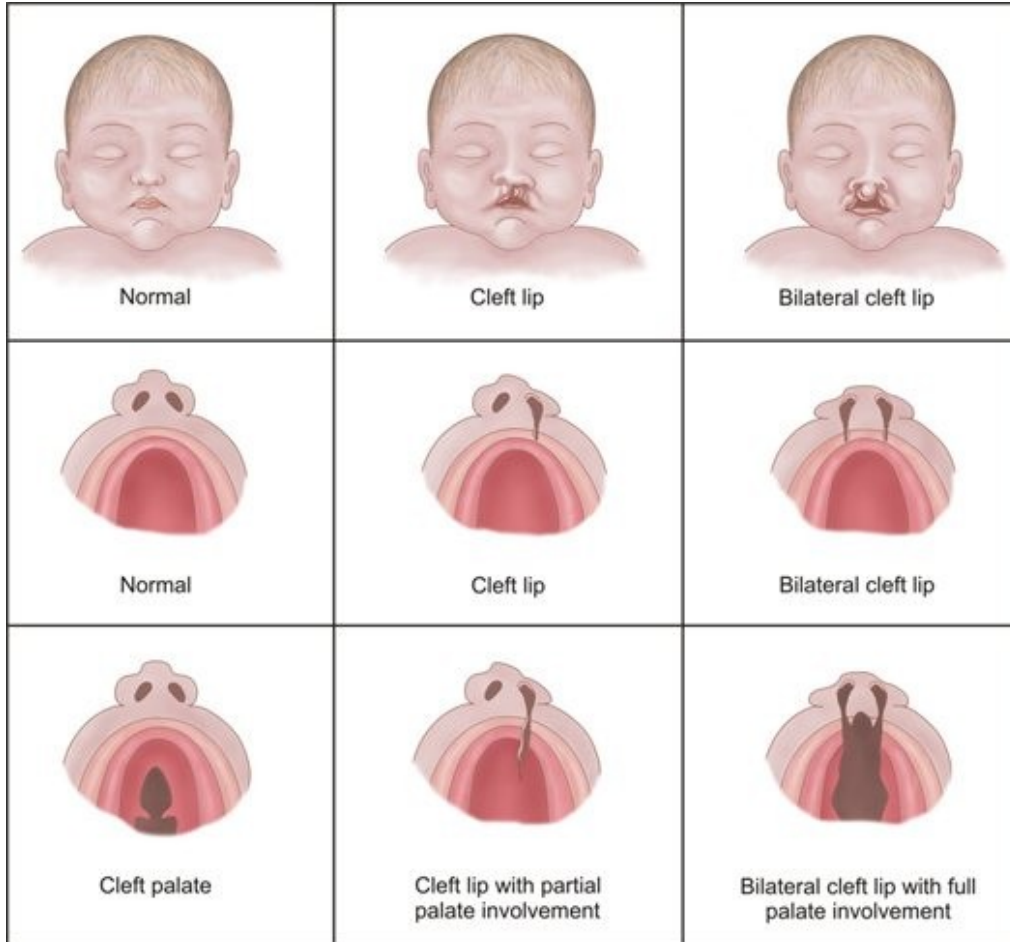
GI anomalies



Cleft Lip and Palate:

Clinical features

- Functional Impairment
 - Suckling, Swallowing, Speech, Hearing, Malocclusion
- Separation of the lip
- Separation of the palate (roof of the mouth)
- Nasal distortion
- Recurring ear infections
- Failure to gain weight
- Nasal regurgitation when bottle feeding
- Poor speech
- Misaligned teeth
- Growth retardation
- Ear infections
- Hearing loss
- Dental cavities
- Displaced teeth
- Poor speech
- Lip deformities
- Nasal deformities



Treatment

■ Feeding

- use of soft artificial nipples with large openings, a squeezable bottle (*Mead Johnson bottle!*)

■ Surgical closure of the cleft lip

- by 3 mo of age, when the infant has shown satisfactory weight gain
- The initial repair may be revised at 4 or 5 yr of age
- Nasal surgery can also be performed at the time of the lip repair.

■ Cleft palate

- The goals of surgery are the union of the cleft segments, intelligible and pleasant speech, reduction of nasal regurgitation, and avoidance of injury to the growing maxilla
- the timing of surgical correction should be individualized but advised to be done before 1 year to enhance normal speech development

Tracheoesophageal Fistula

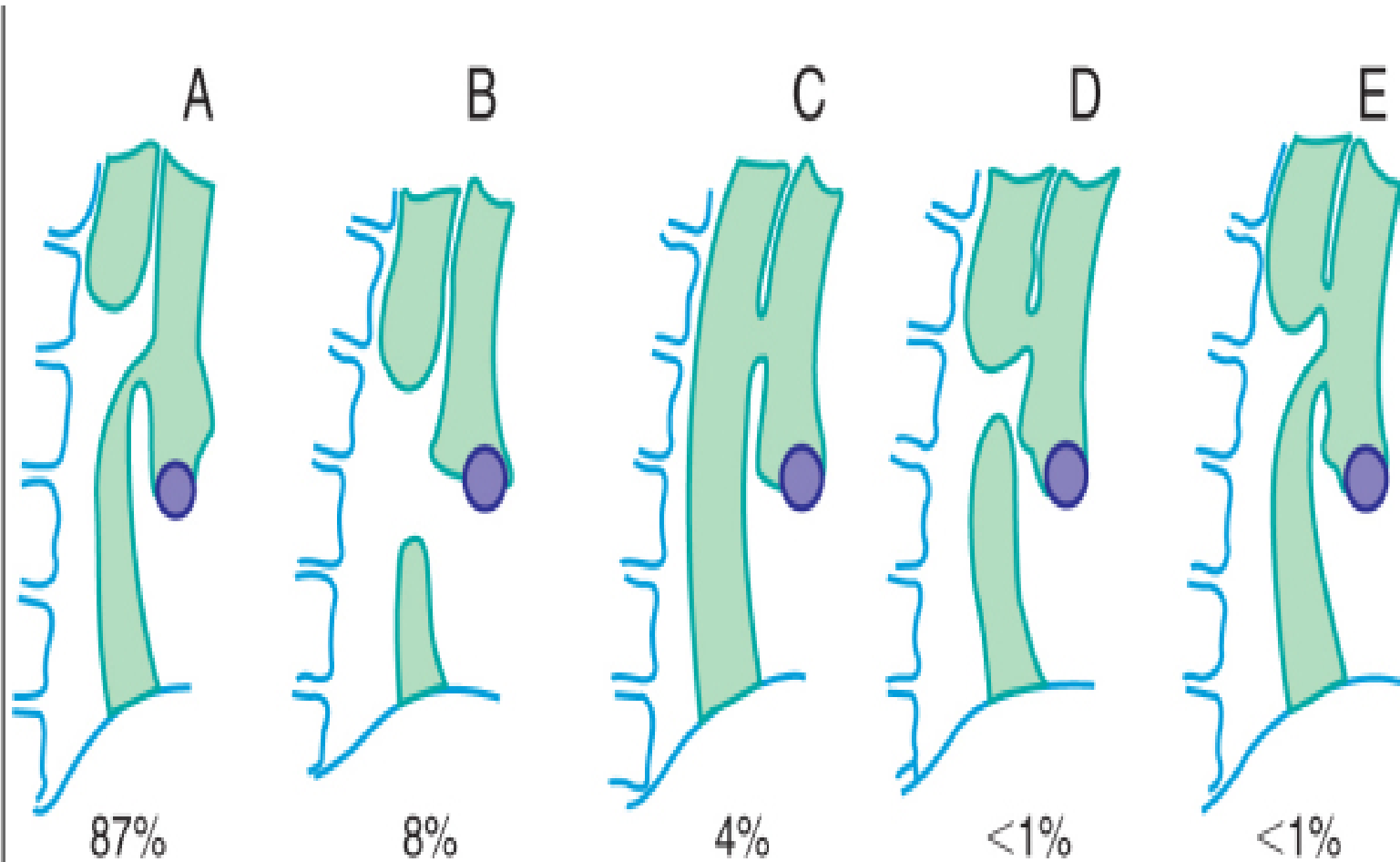
➤ Incidence

- EA and TEF are relatively common congenital anomalies
- ~1 in 4500 live births
- Male=Female
- 0.5% - 2% increase risk in newborns with one affected sibling
- Risk increases to 20% if more than one sibling is affected

Embryologically,

- Derived from primitive foregut
- 4th week of gestation tracheoesophageal diverticulum forms from the laryngotracheal groove
- Tracheoesophageal septum develops during 4th-5th weeks –muscular + submucosal layer of T + E formed
- Elongates with descent of heart and lung
- 7th week reaches final length

Types of tracheoesophageal abn.



Types of tracheoesophageal abn.

- 85% are Distal TEF
- 8% are Pure atresia
- 4% are H-type fistula
- Proximal fistula is the least common

Associated Anomalies

VACTERL: vertebral, anal, cardiac, TE, renal, limb

➤ Congenital heart disease is associated with higher mortality

- VSD is most common
- ASD
- Tetralogy of Fallot
- PDA

➤ Echocardiogram – important to determine position of aortic arch

TE FISTULA/ESOPHAGEAL ATRESIA Associated Defects

Heart defects	59%
Vertebral/rib defects	26%
Kidney defects	16%
Genital defects (males)	15%
Central nervous system defects	13%
Imperforate anus	9%
Limb reduction defects	8%
Duodenal atresia	5%
Bladder/urethral defects	5%
Malrotation of intestines	4%
Genital defects(females)	4%
Hydrocephalus	3%

Source: Torfs et al, 1999

Clinical Features of TEF

▪ Prenatally

- Predictive value of prenatal ultrasound ~20-40%
- Polyhydramnios (2/3 cases)
- Small or absent stomach bubble

▪ Postnatal-

- Most infants are symptomatic within the first few hours of life
- Excessive salivation
- Regurgitation of first feed
- Cyanosis with/without feeds
- Respiratory distress
- Inability to pass a catheter into the stomach
- Gastric distention (with distal fistula)

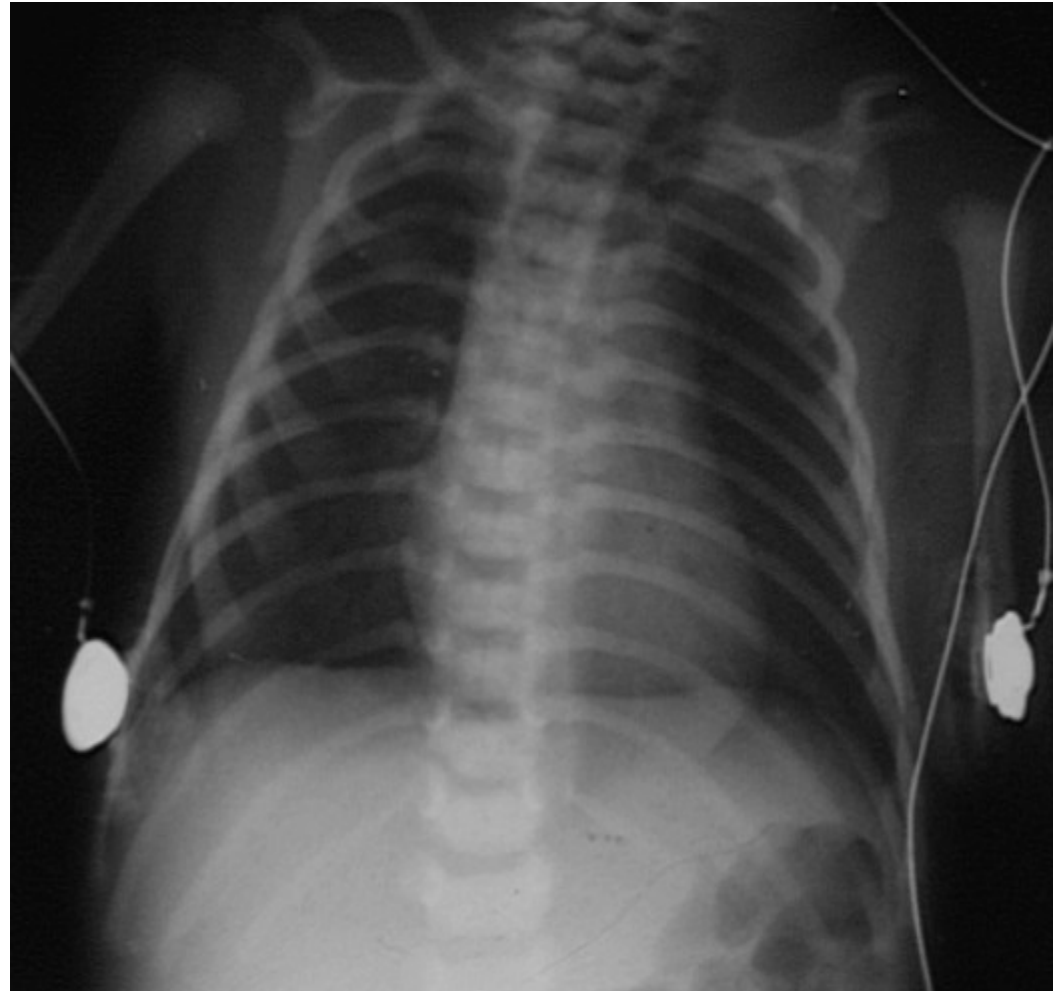
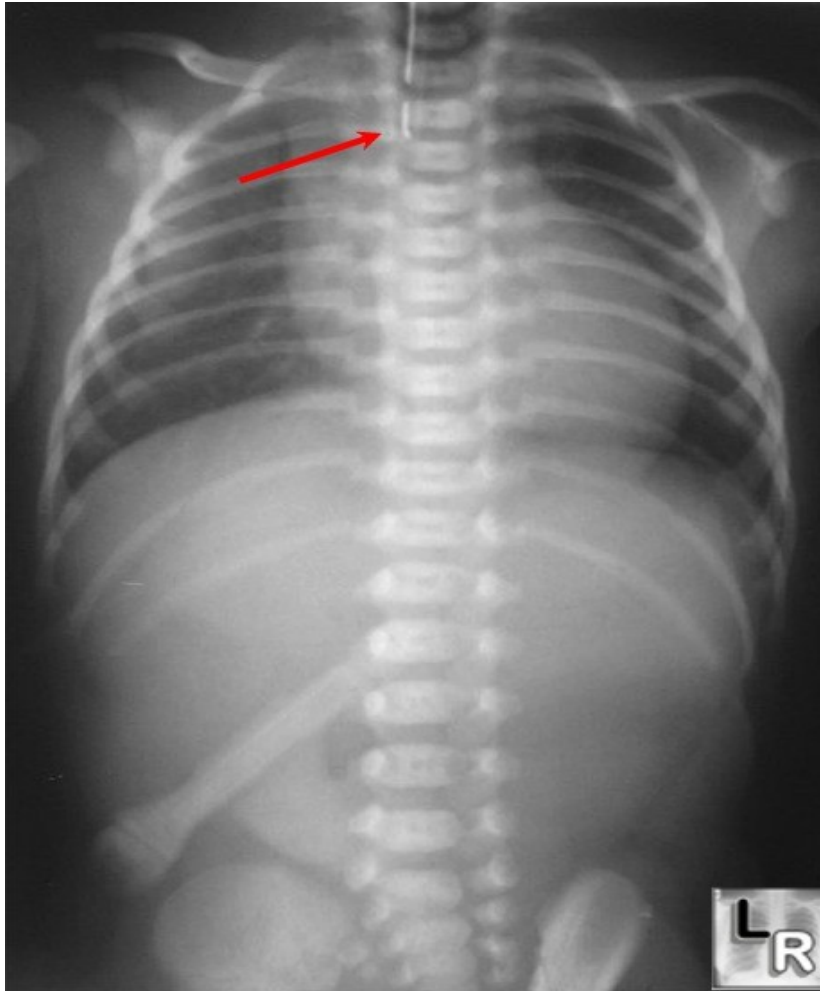
Diagnosis of TEF

- **Failure to pass NG tube (not the H type)**
- **CXR- tube coiled in upper esophagus**
- **“Pouchogram” with air**
- **Distal air on AXR confirms the presence of a fistula**
- **H-type fistulas are often diagnosed later**
 - May present with recurrent pneumonia
 - Delay in diagnosis ranged from 26 days to 4 years.

Diagnosis

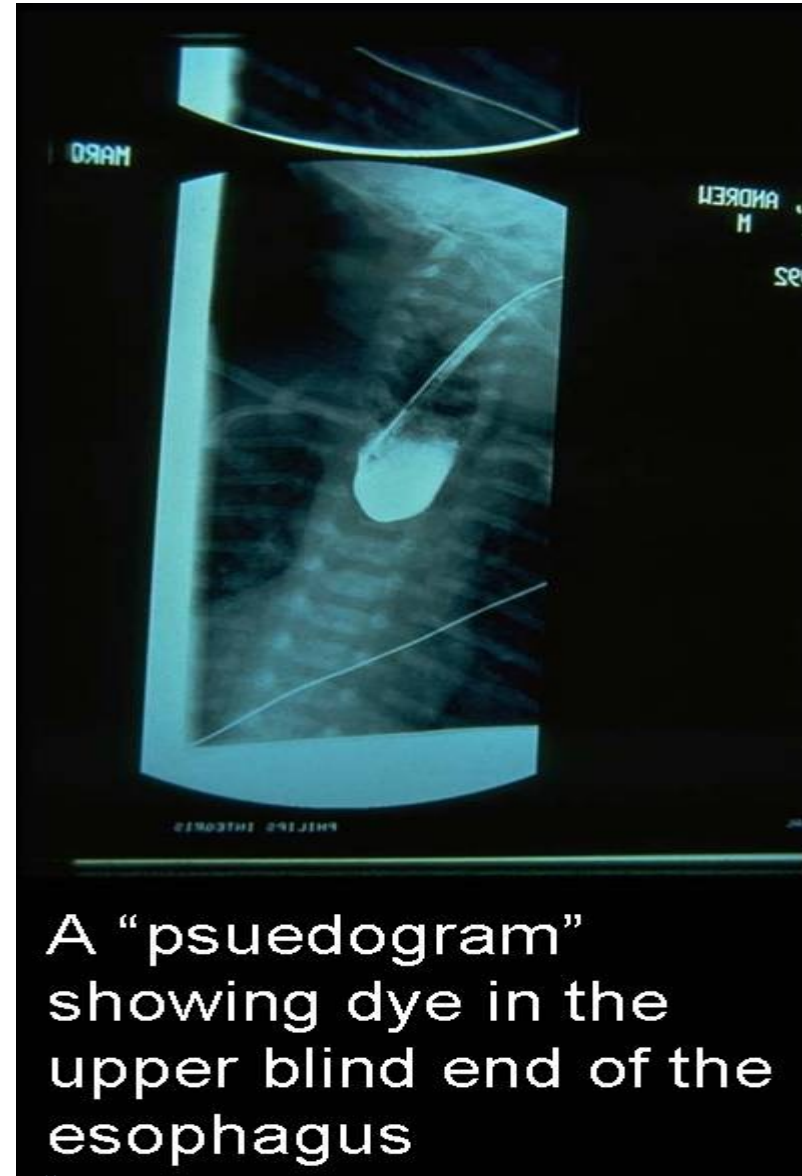
Confirm with AP chest x-ray that demonstrates the catheter curled in the upper esophageal pouch.

- Abdominal XR can help distinguish esophageal blind pouch (no gastric air) from distal TEF (gastric air)



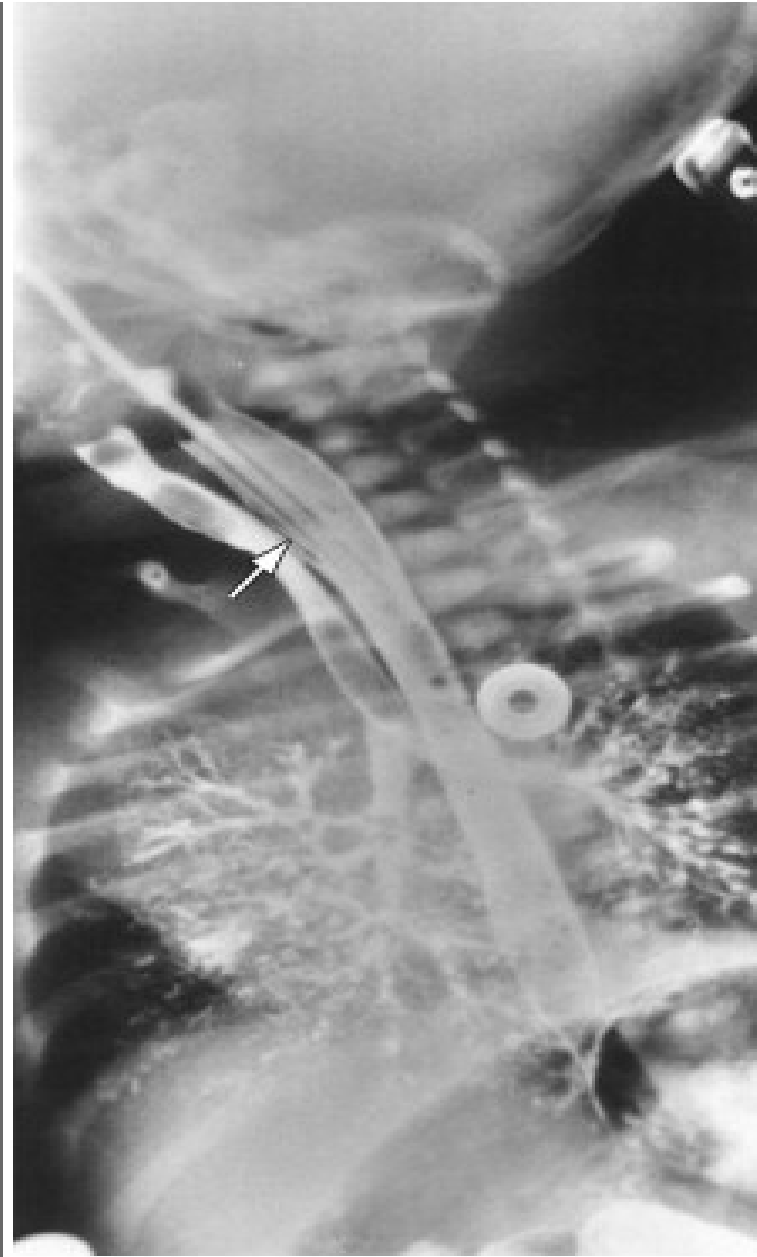
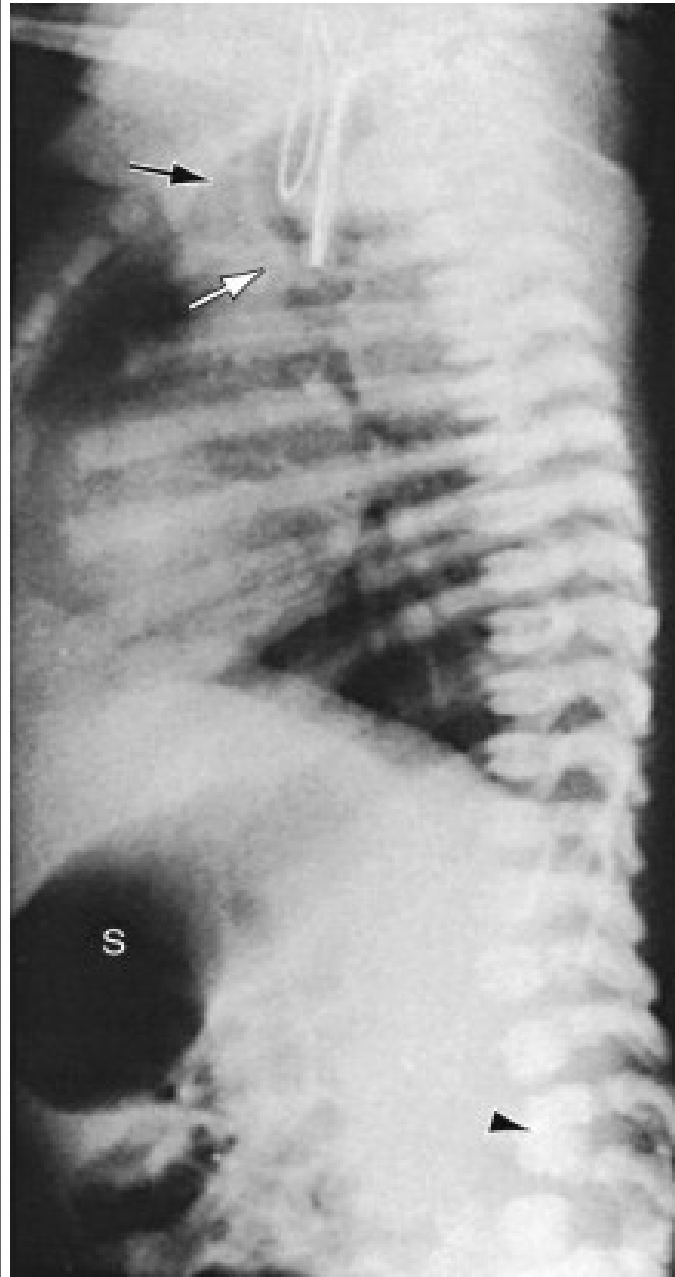
Diagnosis

When diagnosis is uncertain or proximal TEF is suspected, a small amount of water-soluble contrast material can be injected into the esophageal pouch under fluoroscopic guidance (must remove contrast material immediately to avoid regurgitation and aspiration)



Diagnosis- Type E “H type” TEF





Trachioesophageal disorder

➤ Management

- Preoperatively
 - Minimize complications from aspiration
 - Suction blind pouch continuously
 - NPO/total parenteral nutrition/
 - Upright position of child
 - Early surgery for short gap atresia
 - Long gap atresia may be delayed up to 6-12 weeks
 - Long gap atresia should have gastrostomy tube placed for enteral nutrition

▪ Surgery

- Surgical ligation of the fistula
- Primary anastomosis of the esophageal segments
- Primary repair may not be possible if the distance between esophageal segments is long.
- Staged procedures have been performed that include elongation of the esophagus with circular myotomy, interposition of the colon, and gastric transposition

Post op

- Adequate fluid resuscitation
- Wean ventilator
- No deep tracheal suctioning
- Avoid bag/mask ventilation
- No pacifiers/sucking
- Chest tube remains in place for approximately one week
- Esophagram 1 week post op

Prognosis

- 85% to 95% survival
- Infants with increased risk of morbidity and mortality
 - Low birth weight (>1500)
 - Major CHD
 - Severe associated anomalies
 - Ventilator dependency
 - Long gap atresia : >3-4 cm, primary repair cannot be done; options include using gastric, jejunal, or colonic segments interposed as a neo -esophagus

➤ Outcome

▪ Complications

- GER 40-70%
- Esophageal stricture 40%
- Anastomotic leak 14-21%
- tracheomalacia
- fistula recurrence
- esophageal dysmotility Disturbed peristalsis and delayed gastric emptying are common

